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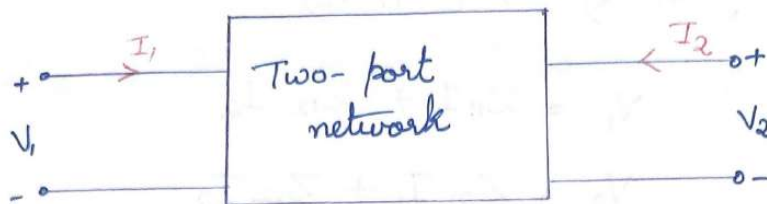
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MODULE- 5

Two port networks : Driving point and transfer functions - Z , Y , h and T parameters - Conditions for symmetry, reciprocity - relationship between parameter sets - interconnections of two port networks (series, parallel and cascade) - T - π transformation.

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TWO - PORT NETWORKS



A two port network has two pairs of terminals.

One pair at inputs - input port

One pair at output - output port

There are 4 variables - V_1, V_2, I_1 and I_2

There will be 2 dependent and 2 independent variables

Parameter	Variables		Equation
	Express	in terms of	
Open Circuit impedance	V_1, V_2	I_1, I_2	$V_1 = Z_{11} I_1 + Z_{12} I_2$ $V_2 = Z_{21} I_1 + Z_{22} I_2$
Short Circuit admittance	I_1, I_2	V_1, V_2	$I_1 = Y_{11} V_1 + Y_{12} V_2$ $I_2 = Y_{21} V_1 + Y_{22} V_2$
Transmission	V_1, I_1	V_2, I_2	$V_1 = A V_2 - B I_2$ $I_1 = C V_2 - D I_2$
Inverse transmission	V_2, I_2	V_1, I_1	$V_2 = A' V_1 - B' I_1$ $I_2 = C' V_1 - D' I_1$
Hybrid	V_1, I_2	V_2, I_1	$V_1 = h_{11} I_1 + h_{12} V_2$ $I_2 = h_{21} I_1 + h_{22} V_2$

OPEN CIRCUIT IMPEDANCE PARAMETERS (Z-PARAMETERS)

$$(V_1, V_2) = f(I_1, I_2)$$

$$V_1 = Z_{11} I_1 + Z_{12} I_2$$

$$V_2 = Z_{21} I_1 + Z_{22} I_2$$

$$\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$$

$$\underline{V = [Z] I}$$

$$\underline{Z_{11} = \frac{V_1}{I_1} \Big|_{I_2=0}} ; I_2=0 \Rightarrow \text{o/p port is open Circuited.}$$

Z_{11} is Driving point impedance with o/p open circuited
or
Open Circuit input impedance.

$$\underline{Z_{21} = \frac{V_2}{I_1} \Big|_{I_2=0}}$$

Z_{21} is Transfer impedance with o/p port open circuited.
or
Open Circuit forward transfer impedance

$$\underline{Z_{12} = \frac{V_1}{I_2} \Big|_{I_1=0}}$$

$I_1=0 \Rightarrow$ i/p port is open circuited.

Z_{12} is the transfer impedance with the i/p port open circuited.

or

Open Circuit reverse transfer impedance.

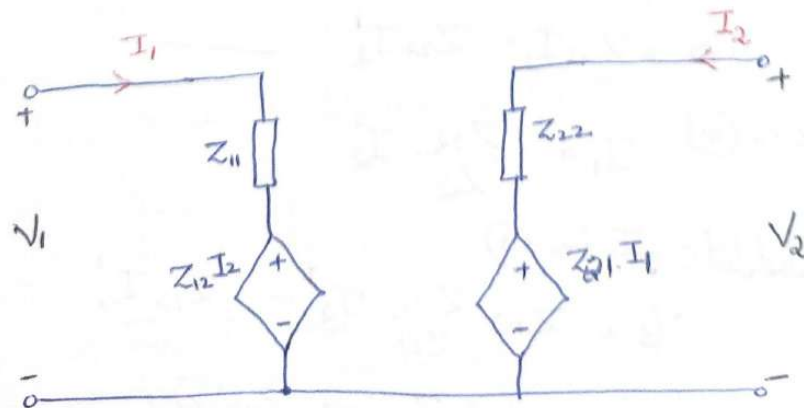
$$\underline{Z_{22} = \frac{V_2}{I_2} \Big|_{I_1=0}}$$

Z_{22} is the open circuit Driving point impedance with the i/p port open circuited.

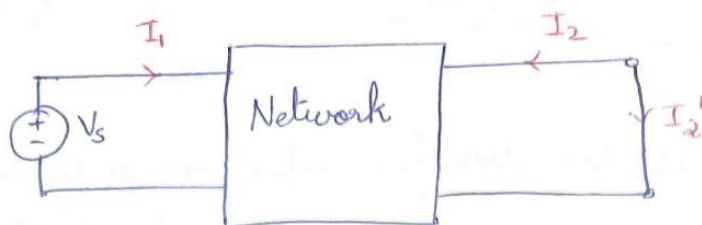
or

Open circuit output impedance.

Equivalent circuit of the 2port network



Condition for Reciprocity



A network is said to be reciprocal if the ratio of the excitation at one port to response at other port is same if excitation and response are interchanged.

From Fig; $V_1 = V_s$

$$V_2 = 0$$

$$I_2 = -I_2'$$

From Z-parameter equations,

$$V_1 = Z_{11} I_1 + Z_{12} I_2$$

$$V_s = Z_{11} I_1 - Z_{12} I_2' \quad \text{--- (1)}$$

and $V_2 = Z_{21} I_1 + Z_{22} I_2$

$$0 = Z_{21} I_1 - Z_{22} I_2' \quad \text{--- (2)}$$

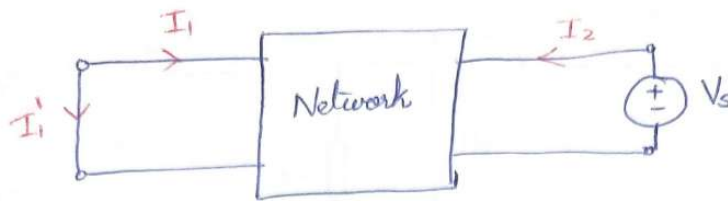
From (2) $I_1 = \frac{Z_{22}}{Z_{21}} I_2'$

Substitute I_1 in (1)

$$V_s = Z_{11} \cdot \frac{Z_{22}}{Z_{21}} I_2' - Z_{12} I_2'$$

$$\frac{V_s}{I_2'} = \frac{Z_{11} Z_{22} - Z_{12} Z_{21}}{Z_{21}} \quad \text{--- (A)}$$

Excitation and Response are interchanged



$$V_2 = V_s ; \quad V_1 = 0$$

$$I_1 = -I_1'$$

$$V_1 = Z_{11} I_1 + Z_{12} I_2$$

$$V_2 = Z_{21} I_1 + Z_{22} I_2$$

Z-parameter equations

$$0 = -Z_{11} I_1' + Z_{12} I_2 \quad \text{--- (3)}$$

$$V_s = -Z_{21} I_1' + Z_{22} I_2 \quad \text{--- (4)}$$

$$\textcircled{3} \Rightarrow I_2 = \frac{Z_{11}}{Z_{12}} I_1'$$

Substitute in (4)

$$V_s = -Z_{21} I_1' + Z_{22} \cdot \frac{Z_{11}}{Z_{12}} I_1'$$

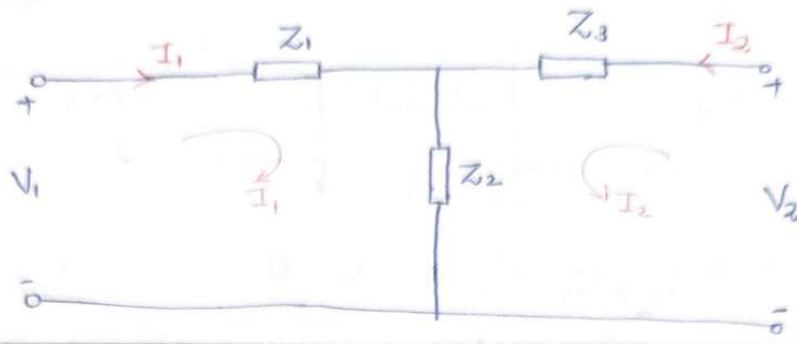
$$\frac{V_s}{I_1'} = \frac{Z_{11} Z_{22} - Z_{12} Z_{21}}{Z_{12}} \quad \text{--- (5)}$$

For the network to be reciprocal,

$$\frac{V_s}{I_1'} = \frac{V_s}{I_2'}$$

$$\underline{\underline{Z_{12} = Z_{21}}}$$

1. Find the Z-parameters for the network shown



Applying KVL to mesh 1

$$V_1 = Z_1 I_1 + Z_2 (I_1 + I_2)$$

$$V_1 = (Z_1 + Z_2) I_1 + Z_2 I_2 \quad \text{--- (1)}$$

Applying KVL to mesh 2

$$V_2 = Z_3 I_2 + Z_2 (I_1 + I_2)$$

$$V_2 = Z_2 I_1 + (Z_2 + Z_3) I_2 \quad \text{--- (2)}$$

From ① & ②

$$Z_{11} = Z_1 + Z_2$$

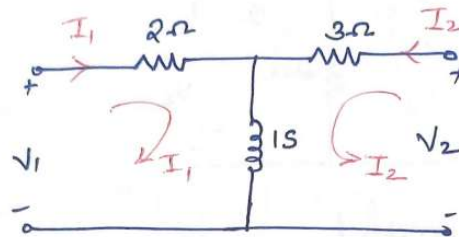
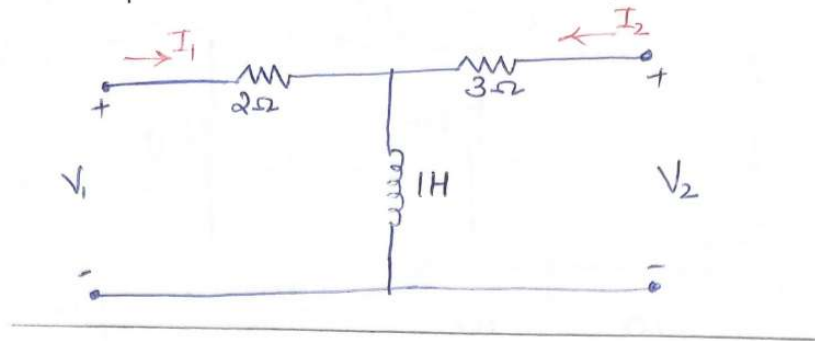
$$Z_{12} = Z_2$$

$$Z_{21} = Z_2$$

$$Z_{22} = Z_2 + Z_3$$

$$\begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} = \begin{bmatrix} Z_1 + Z_2 & Z_2 \\ Z_2 & Z_2 + Z_3 \end{bmatrix}$$

2. Find Z-parameters of the two port circuit.



Apply KVL around the two loops

$$V_1 = 2I_1 + 1S(I_1 + I_2)$$

$$V_1 = (2+S)I_1 + S I_2 \quad \text{--- (1)}$$

$$V_2 = 3I_2 + 1S(I_1 + I_2)$$

$$V_2 = S I_1 + (3+S)I_2 \quad \text{--- (2)}$$

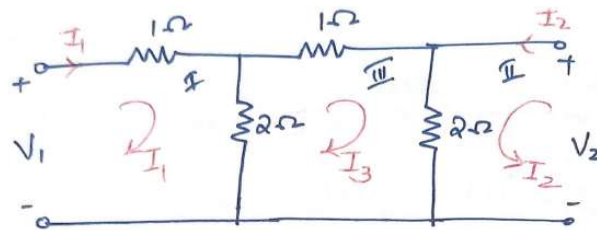
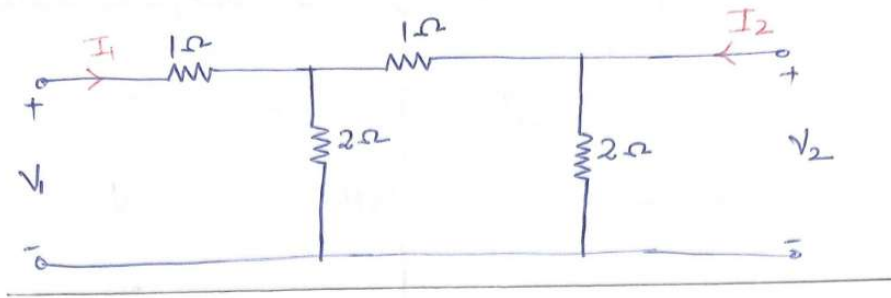
By Comparing (1) & (2) with general eqn $\rightarrow \begin{cases} V_1 = Z_{11}I_1 + Z_{12}I_2 \\ V_2 = Z_{21}I_1 + Z_{22}I_2 \end{cases}$

$$Z_{11} = 2+S$$

$$Z_{12} = Z_{21} = S$$

$$Z_{22} = S+3$$

3. Find Z-parameters for the networks shown.



Applying KVL to mesh 1

$$V_1 = 1 I_1 + 2(I_1 + I_3)$$

$$V_1 = 3I_1 - 2I_3 \quad \text{--- (1)}$$

Applying KVL to mesh 2

$$V_2 = 2(I_2 + I_3)$$

$$V_2 = 2I_2 + 2I_3 \quad \text{--- (2)}$$

Applying KVL to mesh 3

$$2(I_3 - I_1) + I_3 + 2(I_3 + I_2) = 0$$

$$-2I_1 + 2I_2 + 5I_3 = 0 \quad \text{--- (3)}$$

$$\textcircled{3} \Rightarrow I_3 = \frac{2}{5} I_1 - \frac{2}{5} I_2 \quad \text{--- (4)}$$

Substituting (4) in (1)

$$V_1 = 3I_1 - 2 \left\{ \frac{2}{5} I_1 - \frac{2}{5} I_2 \right\}$$

$$V_1 = \frac{11}{5} I_1 + \frac{4}{5} I_2 \quad \text{--- (A)}$$

Substituting (4) in (2)

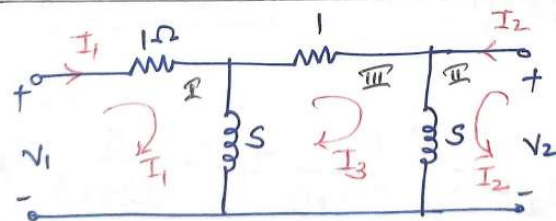
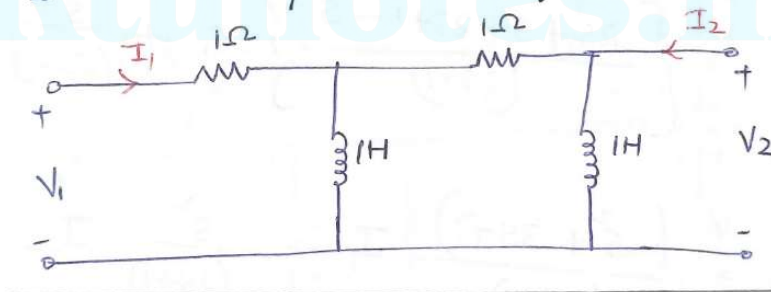
$$V_2 = 2I_2 + 2 \left\{ \frac{2}{5} I_1 - \frac{2}{5} I_2 \right\}$$

$$V_2 = \frac{4}{5} I_1 + \frac{6}{5} I_2 \quad \text{--- (B)}$$

Comparing (A) & (B) with z-parameter equations

$$\begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} = \begin{bmatrix} \frac{11}{5} & \frac{4}{5} \\ \frac{4}{5} & \frac{6}{5} \end{bmatrix}$$

4. Find the z-parameters for the network shown



KVL to mesh 1; $V_1 = I_1 + S(I_1 - I_3)$
 $V_1 = (S+1)I_1 - SI_3 \quad \text{--- (1)}$

KVL to mesh 2; $V_2 = S(I_2 + I_3)$
 $V_2 = SI_2 + SI_3 \quad \text{--- (2)}$

KVL to mesh 3

$$s(I_3 - I_1) + 1I_3 + s(I_3 + I_2) = 0$$

$$-sI_1 + sI_2 + (2s+1)I_3 = 0 \quad \text{--- (3)}$$

$$\textcircled{3} \Rightarrow I_3 = \frac{s}{(2s+1)} I_1 - \frac{s}{(2s+1)} I_2 \quad \text{--- (4)}$$

④ in ①

$$V_1 = (s+1)I_1 - s \left\{ \frac{s}{(2s+1)} I_1 - \frac{s}{(2s+1)} I_2 \right\}$$

$$= \left\{ (s+1) - \frac{s^2}{(2s+1)} \right\} I_1 + \frac{s^2}{(2s+1)} I_2$$

$$= \left\{ \frac{2s^2 + s + 2s + 1 - s^2}{(2s+1)} \right\} I_1 + \frac{s^2}{(2s+1)} I_2$$

$$V_1 = \frac{(s^2 + 3s + 1)}{(2s+1)} I_1 - \frac{s^2}{(2s+1)} I_2 \quad \text{--- (A)}$$

④ in ②

$$\Rightarrow V_2 = sI_2 + s \left\{ \frac{s}{(2s+1)} I_1 - \frac{s}{(2s+1)} I_2 \right\}$$

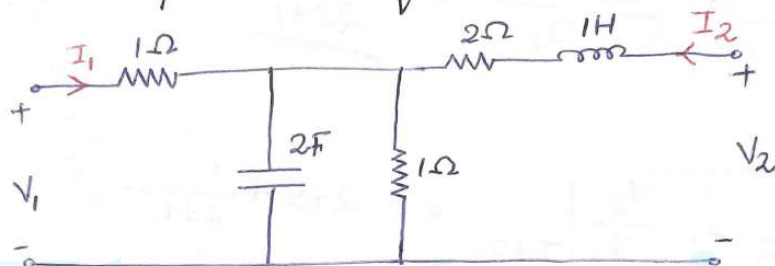
$$= \left\{ s - \frac{s^2}{2s+1} \right\} I_2 + \frac{s^2}{(2s+1)} I_1$$

$$V_2 = \frac{s^2}{(2s+1)} I_1 + \frac{s^2 + s}{(2s+1)} I_2 \quad \text{--- (B)}$$

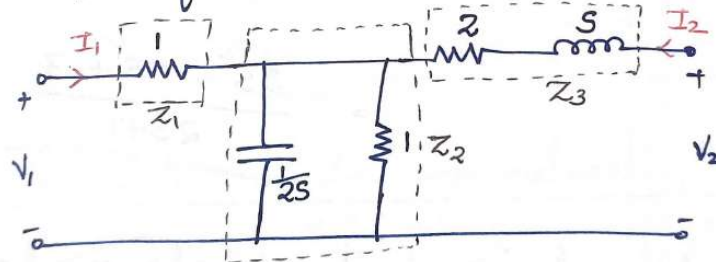
Comparing (A) & (B) with Z-parameter equations

$$\begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} = \begin{bmatrix} \frac{S^2 + 3S + 1}{2S + 1} & \frac{S^2}{2S + 1} \\ \frac{S^2}{2S + 1} & \frac{S^2 + S}{2S + 1} \end{bmatrix}$$

5. Find Z-parameter for the network shown



The transformed network is



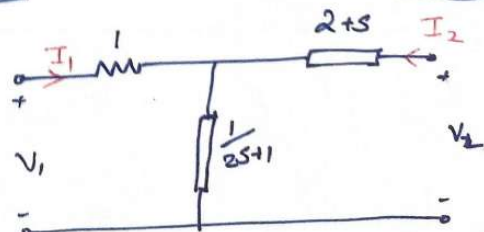
$$Z_1 = 1$$

$$Z_2 = \frac{\left(\frac{1}{2S}\right) \times 1}{\frac{1}{2S} + 1} = \frac{1}{2S + 1}$$

$$Z_3 = S + 2$$

$$Z_{11} = \left. \frac{V_1}{I_1} \right|_{I_2=0} = 1 + \frac{1}{2S + 1} = \frac{2S + 2}{2S + 1}$$

$$\begin{aligned} Z_{11} &= Z_1 + Z_2 \\ Z_{12} &= Z_2 \\ Z_{21} &= Z_2 \\ Z_{22} &= Z_2 + Z_3 \end{aligned}$$



$$Z_{12} = \left. \frac{V_1}{I_2} \right|_{I_1=0}$$

$$= \frac{\frac{1}{2s+1} \cdot I_2}{I_2} = \underline{\underline{\frac{1}{2s+1}}}$$

$$Z_{21} = \left. \frac{V_2}{I_1} \right|_{I_2=0}$$

$$= \frac{\frac{1}{2s+1} \cdot I_1}{I_1} = \underline{\underline{\frac{1}{2s+1}}}$$

$$Z_{22} = \left. \frac{V_2}{I_2} \right|_{I_1=0} = 2+s + \frac{1}{2s+1}$$

$$= \frac{\cancel{1}s + 2s^2 + 2+s+1}{2s+1}$$

$$= \underline{\underline{\frac{2s^2 + 5s + 3}{2s+1}}}$$

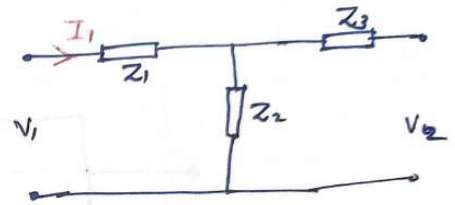
Condition for Symmetry

For a network to be symmetrical, the voltage-to-current ratio at one port should be the same as the voltage-to-current ratio at the other port with one of the ports open-circuited.

a) when o/p port is open circuited i.e. $I_2 = 0$

$$V_1 = Z_{11} I_1$$

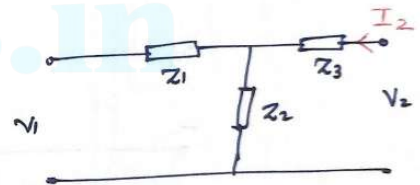
$$\underline{\underline{\frac{V_1}{I_1} = Z_{11}}}$$



b) When i/p port is open circuited; i.e. $I_1 = 0$

$$V_2 = Z_{22} I_2$$

$$\underline{\underline{\frac{V_2}{I_2} = Z_{22}}}$$

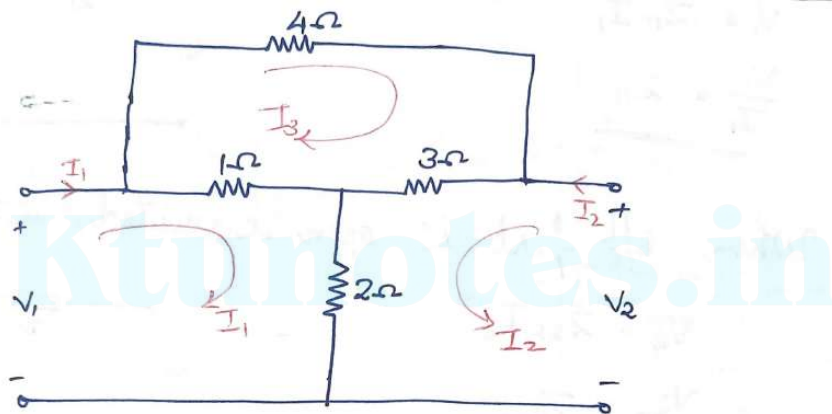
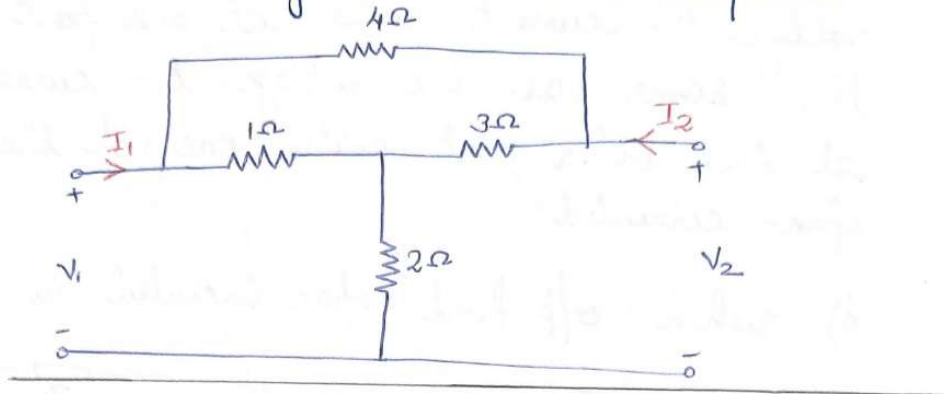


For network to be symmetrical,

$$\frac{V_1}{I_1} = \frac{V_2}{I_2}$$

$$\underline{\underline{Z_{11} = Z_{22}}}$$

6. Find the open circuit impedance parameters for the network shown. Determine whether the network is symmetrical and reciprocal.



Applying KVL to Mesh 1

$$V_1 - 1(I_1 - I_3) - 2(I_1 + I_2) = 0$$

$$V_1 = 3I_1 + 2I_2 - I_3 \quad \text{--- (1)}$$

Applying KVL to mesh 2

$$V_2 - 3(I_2 + I_3) - 2(I_2 + I_1) = 0$$

$$V_2 = 2I_1 + 5I_2 + 3I_3 \quad \text{--- (2)}$$

Applying KVL to mesh 3

$$-4I_3 - 3(I_3 + I_2) - 1(I_3 - I_1) = 0$$

$$I_1 - 3I_2 - 8I_3 = 0 \quad \text{--- (3)}$$

$$(3) \Rightarrow 8I_3 = I_1 - 3I_2$$

$$I_3 = \frac{1}{8}I_1 - \frac{3}{8}I_2 \quad \text{————— (4)}$$

Substituting (4) in (1)

$$V_1 = 3I_1 + 2I_2 - I_3$$

$$V_1 = 3I_1 + 2I_2 - \left\{ \frac{1}{8}I_1 - \frac{3}{8}I_2 \right\}$$

$$= \left(3 - \frac{1}{8}\right)I_1 + \left(2 + \frac{3}{8}\right)I_2$$

$$V_1 = \underline{\underline{\frac{23}{8}I_1 + \frac{19}{8}I_2}} \quad \text{————— (5)}$$

Substituting (4) in (2)

$$V_2 = 2I_1 + 5I_2 + 3I_3$$

$$V_2 = 2I_1 + 5I_2 + 3\left\{ \frac{1}{8}I_1 - \frac{3}{8}I_2 \right\}$$

$$= \left(2 + \frac{3}{8}\right)I_1 + \left(5 - \frac{9}{8}\right)I_2$$

$$V_2 = \underline{\underline{\frac{19}{8}I_1 + \frac{31}{8}I_2}} \quad \text{————— (6)}$$

Z-parameter eqns are

$$V_1 = Z_{11}I_1 + Z_{12}I_2$$

$$V_2 = Z_{21}I_1 + Z_{22}I_2$$

Comparing (5) & (6) with z-parameter equations

$$Z_{11} = \frac{23}{8}$$

$$Z_{12} = \frac{19}{8}$$

$$Z_{21} = \frac{19}{8}$$

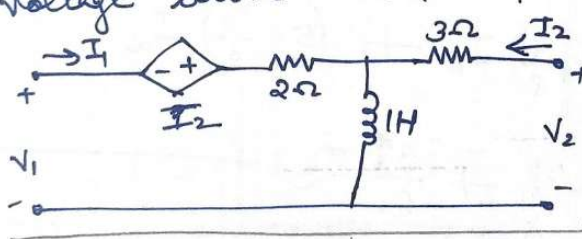
$$Z_{22} = \frac{31}{8}$$

$$\begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} = \begin{bmatrix} \frac{23}{8} & \frac{19}{8} \\ \frac{19}{8} & \frac{31}{8} \end{bmatrix}$$

Since $Z_{11} \neq Z_{22}$; the network is not symmetrical.

Since $Z_{12} = Z_{21}$; the network is reciprocal.

7. The 2port network shown contains a current-dependent voltage source. Find z parameters.



$$\text{loop}^2 \quad \begin{aligned} V_2 &= 3I_2 + 1S(I_2 + I_1) \\ V_2 &= SI_1 + (3+S)I_2 \end{aligned}$$

So z parameters are

$$Z_{11} = (2+S)$$

$$Z_{12} = (S-1)$$

$$Z_{21} = S$$

$$Z_{22} = (3+S)$$

loop 1

$$V_1 + I_2 - 2I_1 - 1S(I_1 + I_2) = 0$$

$$V_1 = (2+S)I_1 + (S-1)I_2$$

SHORT CIRCUIT ADMITTANCE PARAMETERS (Y-PARAMETERS)

Two port currents I_1 & I_2 in terms of the two-port voltages V_1 & V_2 .

$$I_1 = Y_{11} V_1 + Y_{12} V_2$$

$$I_2 = Y_{21} V_1 + Y_{22} V_2$$

$$\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix}$$

$$\underline{[I] = [Y] [V]}$$

$$Y_{11} = \frac{I_1}{V_1} \bigg|_{V_2=0} \quad V_2=0 \Rightarrow \text{o/p port is short circuited}$$

$$Y_{21} = \frac{I_2}{V_1} \bigg|_{V_2=0}$$

Y_{11} is the driving point admittance with the output port short circuited. It is also called short circuit input admittance.

Y_{21} is the transfer admittance with the output port short circuited. It is also called short circuit ~~input admittance~~ forward transfer admittance.

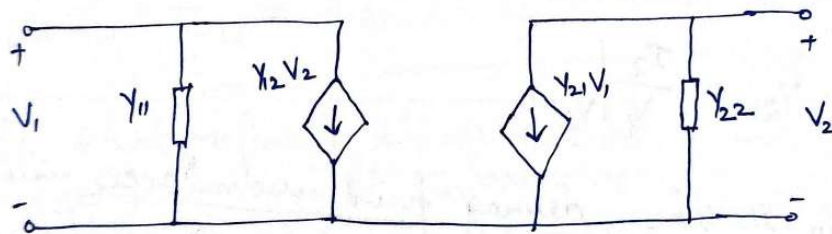
$$Y_{12} = \frac{I_1}{V_2} \Big|_{V_1=0}$$

$V_1=0 \Rightarrow$ i/p port is short circuited.

Y_{12} is the transfer admittance with the ~~input~~ⁱⁿ port short circuited. It is also called short circuit ~~forward~~^{reverse} transfer admittance.

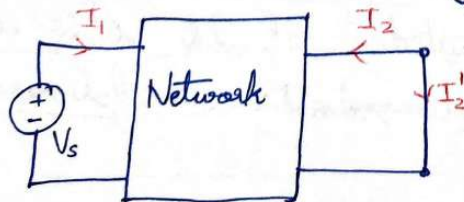
$$Y_{22} = \frac{I_2}{V_2} \Big|_{V_1=0}$$

Y_{22} is the short circuit driving point admittance with the input port short circuited. It is also called the short circuit output admittance.



Equivalent circuit of the 2port n/w in terms of Y parameters.

Condition For Reciprocity



From Y parameter eqn

$$-I_2' = Y_{21} V_s$$

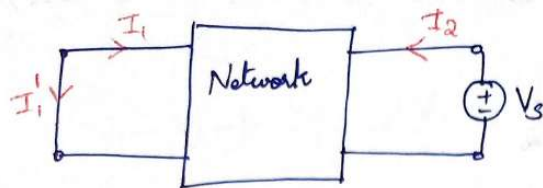
$$\frac{I_2'}{V_s} = -Y_{21}$$

Voltage V_s is applied at input port with the o/p port short circuited

$$V_1 = V_s$$

$$V_2 = 0$$

$$I_2 = -I_2'$$



V_s is applied at o/p port
with the i/p port
short circuited.

$$V_2 = V_s$$

$$V_1 = 0$$

$$I_1 = -I_1'$$

From γ -parameter equation

$$-I_1' = \gamma_{12} V_s$$

$$\underline{\underline{\frac{I_1'}{V_s} = -\gamma_{12}}}$$

For the network to be reciprocal

$$\frac{I_2'}{V_s} = \frac{I_1'}{V_s}$$

$$\text{ie; } \underline{\underline{\gamma_{12} = \gamma_{21}}}$$

Condition for Symmetry

a) when o/p port is short circuited; $V_2 = 0$

$$I_1 = \gamma_{11} V_s \quad \text{from } \gamma \text{ parameter eqn}$$

$$\underline{\underline{\frac{V_s}{I_1} = \gamma_{11}}}$$

b) when i/p port is short circuited; $V_1 = 0$

$$I_2 = \gamma_{22} V_s \quad \text{from } \gamma \text{ parameter eqn}$$

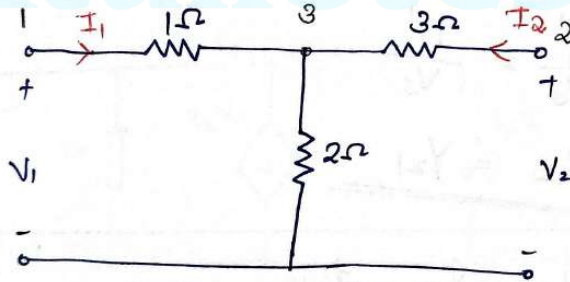
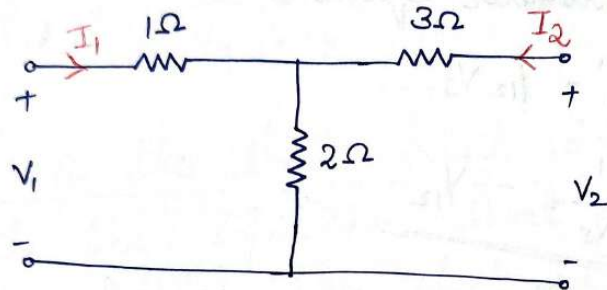
$$\underline{\underline{\frac{V_s}{I_2} = \gamma_{22}}}$$

Hence for the network to be symmetrical,

$$\frac{V_1}{I_1} = \frac{V_2}{I_2}$$

$$\underline{\underline{Y_{11} = Y_{22}}}$$

1. Find the Y -parameters of the network in fig.



$$I_1 = \frac{V_1 - V_3}{1} \quad \text{ie; } I_1 = V_1 - V_3 \quad \text{--- (1)}$$

$$I_2 = \frac{V_2 - V_3}{3} \quad \text{ie; } I_2 = \frac{V_2}{3} - \frac{V_3}{3} \quad \text{--- (2)}$$

Applying KCL at node 3

$$I_1 + I_2 = \frac{V_3}{2} \quad \text{--- (3)}$$

Substituting (1) & (2) in (3)

$$V_1 - V_3 + \frac{V_2}{3} - \frac{V_3}{3} = \frac{V_3}{2}$$

$$V_1 + \frac{V_2}{3} = \frac{11}{6} V_3$$

$$V_3 = \frac{6}{11} V_1 + \frac{2}{11} V_2 \quad \text{--- (4)}$$

Substituting (4) in (1)

$$(1) \Rightarrow I_1 = V_1 - V_3$$

$$(4) \Rightarrow I_1 = \cancel{\frac{6}{11} V_1} - \cancel{\frac{2}{11} V_2} - \cancel{V_1}$$

$$I_1 = V_1 - \frac{6}{11} V_1 - \frac{2}{11} V_2$$

$$I_1 = \frac{5}{11} V_1 - \frac{2}{11} V_2 \quad \text{--- (A)}$$

Substituting (4) in (2)

$$I_2 = \frac{V_2}{3} - \frac{V_3}{3}$$

$$I_2 = \frac{V_2}{3} - \frac{1}{3} \left\{ \frac{6}{11} V_1 + \frac{2}{11} V_2 \right\}$$

$$= \frac{V_2}{3} - \frac{2}{11} V_1 - \frac{2}{33} V_2$$

$$I_2 = -\frac{2}{11} V_1 - \frac{9}{33} V_2$$

$$I_2 = -\frac{2}{11} V_1 - \frac{3}{11} V_2 \quad \text{--- (B)}$$

$$\begin{aligned} \frac{1}{3} - \frac{2}{33} \\ \frac{\frac{11}{33} - \frac{2}{33}}{\frac{9}{33} \cdot \frac{3}{11}} \end{aligned}$$

γ - parameter equations are

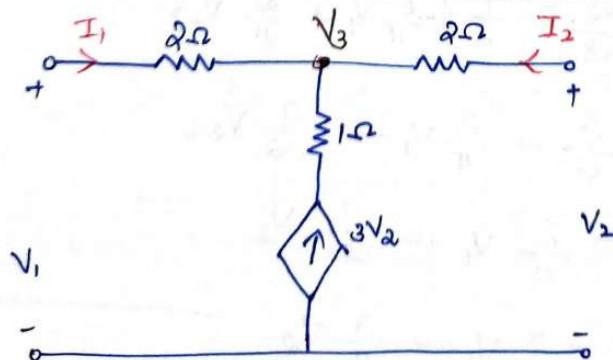
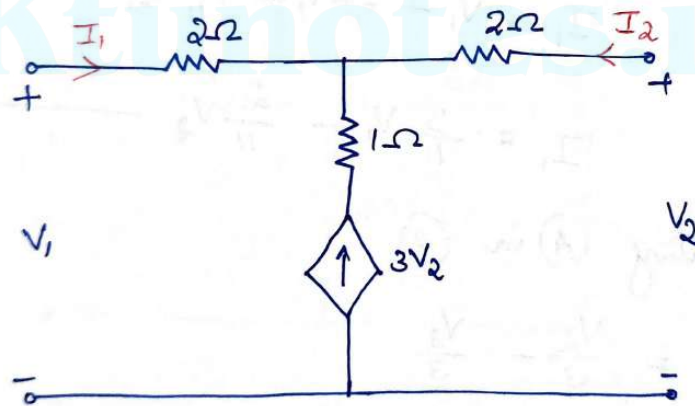
$$I_1 = \gamma_{11} V_1 + \gamma_{12} V_2$$

$$I_2 = \gamma_{21} V_1 + \gamma_{22} V_2$$

Comparing (A) & (B) with γ -parameter equations,

$$\begin{bmatrix} \gamma_{11} & \gamma_{12} \\ \gamma_{21} & \gamma_{22} \end{bmatrix} = \begin{bmatrix} 5/11 & -2/11 \\ -2/11 & -3/11 \end{bmatrix}$$

2. Find γ - parameters of the network shown.



$$I_1 = \frac{V_1 - V_3}{2}$$

$$\underline{I_1 = \frac{1}{2} V_1 - \frac{1}{2} V_3} \quad \text{--- (1)}$$

$$I_2 = \frac{V_2 - V_3}{2}$$

$$\underline{I_2 = \frac{1}{2} V_2 - \frac{1}{2} V_3} \quad \text{--- (2)}$$

Applying KCL at node 3

$$I_1 + I_2 + 3V_2 = 0 \quad \text{--- (3)}$$

Substituting (1) & (2) in (3)

$$\frac{1}{2} V_1 - \frac{1}{2} V_3 + \frac{1}{2} V_2 - \frac{1}{2} V_3 + 3V_2 = 0$$

$$\frac{1}{2} V_1 + \frac{7}{2} V_2 - V_3 = 0$$

$$\underline{V_3 = \frac{1}{2} V_1 + \frac{7}{2} V_2} \quad \text{--- (4)}$$

(4) in (1)

$$\textcircled{1} \Rightarrow I_1 = \frac{1}{2} V_1 - \frac{1}{2} \left\{ \frac{1}{2} V_1 + \frac{7}{2} V_2 \right\}$$

$$= \frac{1}{2} V_1 - \frac{1}{4} V_1 - \frac{7}{4} V_2$$

$$\underline{I_1 = \frac{1}{4} V_1 - \frac{7}{4} V_2} \quad \text{--- (4)}$$

④ in ②

$$I_2 = \frac{1}{2} V_2 - \frac{1}{2} \left\{ \frac{1}{2} V_1 + \frac{7}{2} V_2 \right\}$$

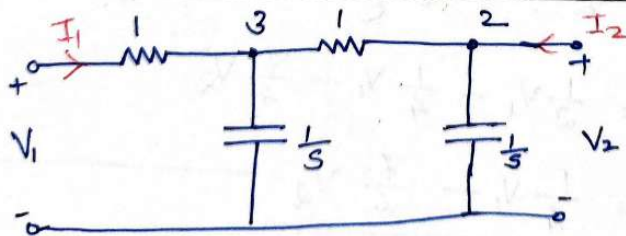
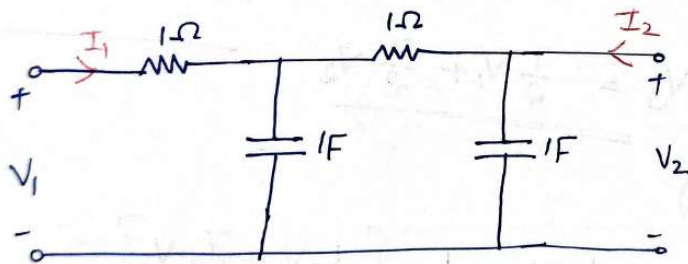
$$I_2 = \frac{1}{2} V_2 - \frac{1}{4} V_1 - \frac{7}{4} V_2$$

$$I_2 = -\frac{1}{4} V_1 - \frac{5}{4} V_2 \quad \text{--- (B)}$$

Comparing ① & ③ with γ -parameter equations

$$\begin{bmatrix} \gamma_{11} & \gamma_{12} \\ \gamma_{21} & \gamma_{22} \end{bmatrix} = \begin{bmatrix} 1/4 & -7/4 \\ -1/4 & -5/4 \end{bmatrix}$$

3. Determine the short circuit admittance parameters



$$I_1 = \frac{V_1 - V_3}{1}$$

$$I_1 = V_1 - V_3 \quad \text{--- (1)}$$

Applying KCL to node 3

$$I_1 = \frac{V_3}{1/s} + \frac{(V_3 - V_2)}{1}$$

$$= sV_3 + V_3 - V_2$$

$$\underline{I_1 = -V_2 + (s+1)V_3} \quad \text{--- (2)}$$

Applying KCL to node 2

$$I_2 = \frac{V_2}{1/s} + \frac{(V_2 - V_3)}{1}$$

$$= sV_2 + (V_2 - V_3)$$

$$\underline{I_2 = (s+1)V_2 - V_3} \quad \text{--- (3)}$$

Substituting ① in ②

$$V_1 - V_3 = -V_2 + (s+1)V_3$$

$$V_1 + V_2 = V_3 + (s+1)V_3$$

$$V_1 + V_2 = (s+2)V_3$$

$$\underline{V_3 = \frac{1}{(s+2)}V_1 + \frac{1}{(s+2)}V_2} \quad \text{--- (4)}$$

Substituting (4) in (2)

$$I_1 = -V_2 + \frac{(s+1)}{(s+2)}V_1 + \frac{(s+1)}{(s+2)}V_2$$

$$\underline{I_1 = \frac{(s+1)}{(s+2)}V_1 - \frac{1}{(s+2)}V_2} \quad \text{--- (A)}$$

Substituting (A) in (3)

$$I_2 = (s+1)V_2 - \frac{1}{(s+2)}V_1 - \frac{1}{(s+2)}V_2$$

$$= -\frac{1}{(s+2)}V_1 + \left\{ (s+1) - \frac{1}{(s+2)} \right\} V_2$$

$$= -\frac{1}{(s+2)}V_1 + \left\{ \frac{s^2+3s+2-1}{(s+2)} \right\} V_2$$

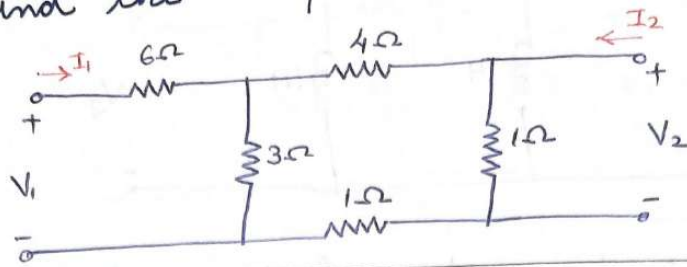
$$= -\frac{1}{(s+2)}V_1 + \frac{s^2+3s+1}{(s+2)}V_2 \quad \text{--- (B')}$$

Comparing (A) & (B') with γ parameter eqns.

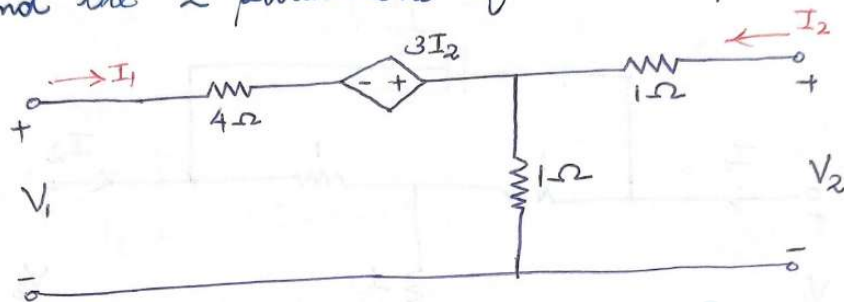
$$\begin{bmatrix} \gamma_{11} & \gamma_{12} \\ \gamma_{21} & \gamma_{22} \end{bmatrix} = \begin{bmatrix} \frac{s+1}{s+2} & -\frac{1}{s+2} \\ -\frac{1}{s+2} & \frac{s^2+3s+1}{s+2} \end{bmatrix}$$

Tutorial Questions

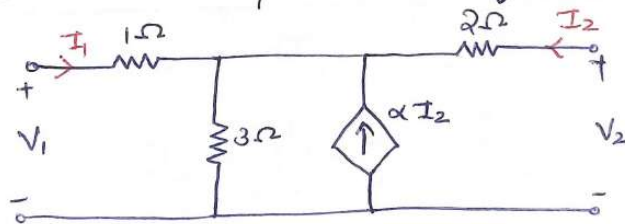
1. Find the Z-parameters



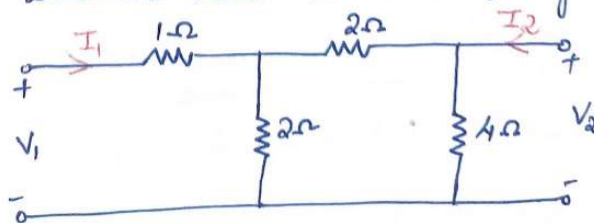
2. Find the Z-parameters of the two port network



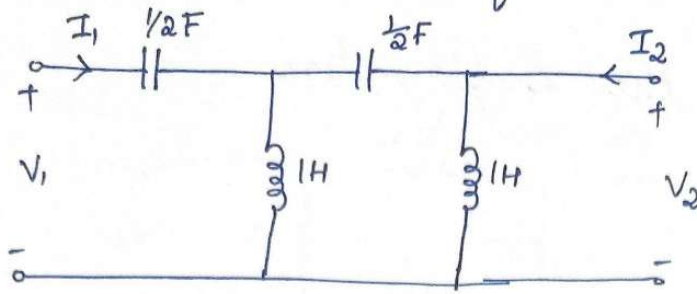
3. Find the Z-parameters of the given network



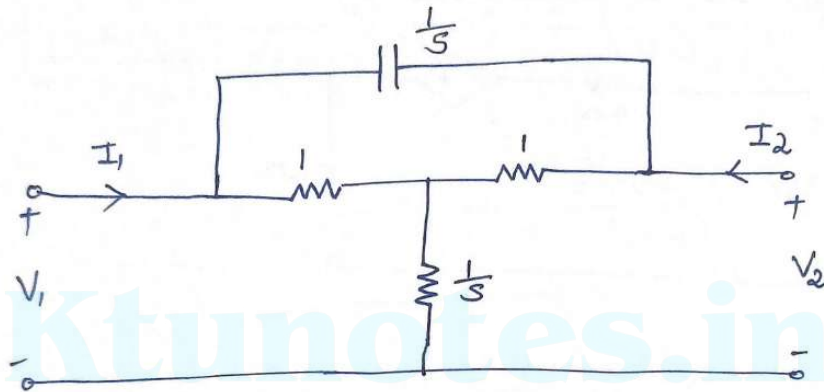
4. Determine Y-parameters for the network shown. Determine whether the network is symmetrical & reciprocal.



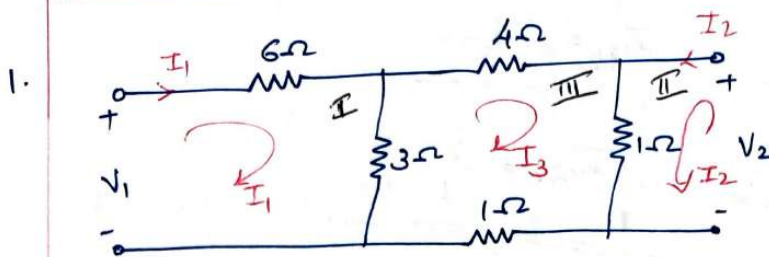
5. Determine Y -parameters for the network shown.



6. Obtain Y -parameters of the network



Solutions for Tutorial



KVL to mesh I

$$V_1 = 6I_1 + 3(I_1 - I_3)$$

$$V_1 = 9I_1 - 3I_3 \quad \text{--- (1)}$$

KVL to mesh II

$$V_2 = 1(I_2 + I_3)$$

$$V_2 = I_2 + I_3 \quad \text{--- (2)}$$

KVL to mesh III

$$-3(I_3 - I_1) - 4I_3 - 1(I_3 + I_2) - I_3 = 0$$

$$3I_1 - I_2 - 9I_3 = 0 \quad \text{--- (3)}$$

$$I_3 = \frac{1}{3}I_1 - \frac{1}{9}I_2 \quad \text{--- (4)}$$

④ in ①

$$V_1 = 9I_1 - 3I_3 = 9I_1 - 3\left\{\frac{1}{3}I_1 - \frac{1}{9}I_2\right\}$$

$$= 9I_1 - I_1 + \frac{1}{3}I_2$$

$$V_1 = 8I_1 + \frac{1}{3}I_2 \quad \text{--- (A)}$$

④ in ②

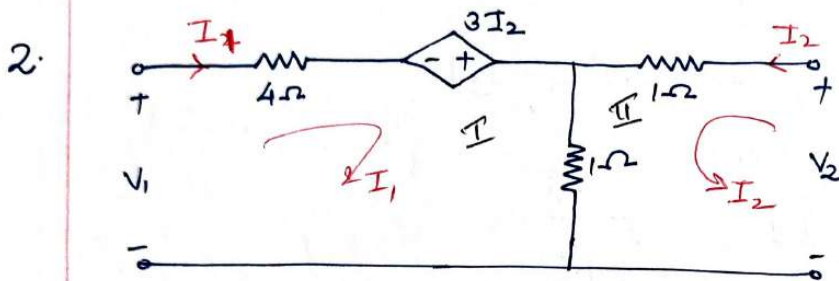
$$V_2 = I_2 + I_3 = I_2 + \frac{1}{3}I_1 - \frac{1}{9}I_2 = \frac{1}{3}I_1 + \left(1 - \frac{1}{9}\right)I_2$$

$$V_2 = \frac{1}{3}I_1 + \frac{8}{9}I_2 \quad \text{--- (B)}$$

Comparing (A) & (B) with Z parameter eqns

$$\underline{Z_{11} = 8} \quad \underline{Z_{12} = \frac{1}{3}}$$

$$\underline{Z_{21} = \frac{1}{3}} \quad \underline{Z_{22} = \frac{8}{9}}$$



KVL to mesh 1

$$V_1 = 4I_1 - 3I_2 + 1(I_1 + I_2)$$

$$\text{ie; } \underline{V_1 = 5I_1 - 2I_2} \quad \text{--- (1)}$$

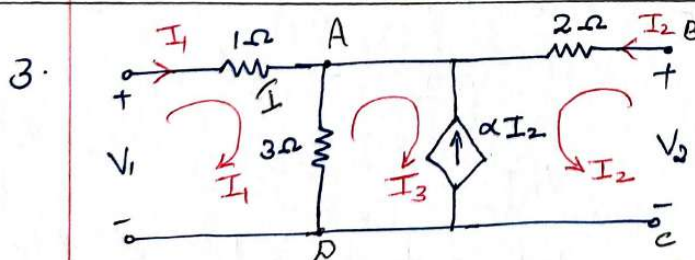
KVL to mesh 2

$$V_2 = 1I_2 + 1(I_2 + I_1)$$

$$\text{ie; } \underline{V_2 = I_1 + 2I_2} \quad \text{--- (2)}$$

Comparing (1) & (2) with Z-parameter eqns

$$\underline{Z_{11} = 5, \quad Z_{12} = -2, \quad Z_{21} = 1, \quad Z_{22} = 2}$$



From the diagram, $I_2 + I_3 = -\alpha I_2$

$$\text{ie; } I_3 = \underline{-(1+\alpha)I_2}$$

$$I_2 = \underline{-(1+\alpha)I_2} \quad \text{--- (1)}$$

Applying KVL to mesh 1 $V_1 - 1I_1 - 3(I_1 - I_3) = 0$

$$V_1 = 1I_1 + 3(I_1 - I_3)$$

$$V_1 = 4I_1 - 3I_3 \quad \text{--- (2)}$$

Applying KVL to loop ABCD $\Rightarrow V_2 - 2I_2 + 3(I_3 - I_1) = 0$

$$V_2 = 2I_2 - 3(I_3 - I_1)$$

$$= 2I_2 - 3I_3 + 3I_1$$

$$\therefore V_2 = 3I_1 + 2I_2 - 3I_3 \quad \text{--- (3)}$$

Substitute (1) in (2)

$$V_1 = 4I_1 - 3I_3$$

$$= 4I_1 - 3(-(1+\alpha)I_2)$$

$$= 4I_1 + 3(1+\alpha)I_2 \quad \text{--- (A)}$$

Substitute (1) in (3)

$$V_2 = 3I_1 + 2I_2 - 3I_3$$

$$= 3I_1 + 2I_2 - 3(1+\alpha)I_2$$

$$= 3I_1 + 2I_2 + 3(1+\alpha)I_2$$

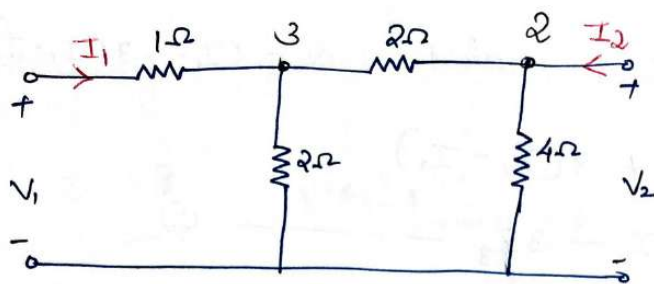
$$= 3I_1 + (5+3\alpha)I_2 \quad \text{--- (B)}$$

Equating (A) & (B) to Z parameter eqns.

$$Z_{11} = 4\Omega; \quad Z_{12} = \underline{\underline{3(1+\alpha)\Omega}}$$

$$Z_{21} = \underline{\underline{3\Omega}} \quad \text{and} \quad Z_{22} = \underline{\underline{(5+3\alpha)\Omega}}$$

4.



$$I_1 = \frac{V_1 - V_3}{1} ; \quad I_1 = \frac{V_1}{1} - \frac{V_3}{1} \quad \text{--- (1)}$$

$$\underline{I_1 = V_1 - V_3} \quad \text{--- (1)}$$

Applying KCL at node 3

$$I_1 = \frac{V_3}{2} + \frac{V_3 - V_2}{2}$$

$$\underline{I_1 = -\frac{V_2}{2} + V_3} \quad \text{--- (2)}$$

Applying KCL at node 2

$$I_2 = \frac{V_2}{4} + \frac{V_2 - V_3}{2}$$

$$\underline{I_2 = \frac{3}{4} V_2 - \frac{V_3}{2}} \quad \text{--- (3)}$$

Substituting (1) in (2)

$$V_1 - V_3 = -\frac{V_2}{2} + V_3$$

$$V_1 + \frac{V_2}{2} = 2V_3$$

$$\underline{V_3 = \frac{1}{2} V_1 + \frac{1}{4} V_2} \quad \text{--- (4)}$$

Substituting (4) in (2)

$$I_1 = -\frac{V_2}{2} + \frac{1}{2} V_1 + \frac{1}{4} V_2$$

$$\underline{I_1 = \frac{1}{2} V_1 - \frac{1}{4} V_2} \quad \text{--- (A)}$$

Substituting (4) in (3)

$$I_2 = \frac{3}{4} V_2 - \frac{1}{2} \left\{ \frac{1}{2} V_1 + \frac{1}{4} V_2 \right\} \quad \left| \begin{array}{l} \frac{3}{4} - \frac{1}{8} \\ = \frac{5}{8} \end{array} \right.$$

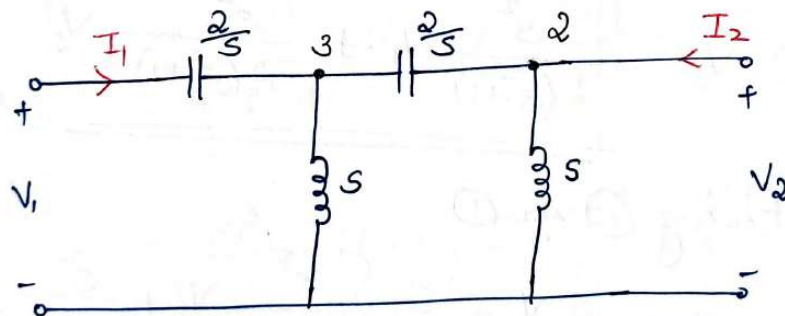
$$I_2 = -\frac{1}{4} V_1 + \frac{5}{8} V_2 \quad \text{--- (C)}$$

~~Substituting~~ Comparing (A) & (C) with Y -parameters

$$\begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & -\frac{1}{4} \\ -\frac{1}{4} & \frac{5}{8} \end{bmatrix}$$

Since $Y_{11} \neq Y_{22}$; Network is not symmetrical
 Since $Y_{12} = Y_{21}$; Network is reciprocal.

5.



$$\frac{1}{CS} = \frac{1}{\frac{1}{2}S} = \frac{2}{S}$$

$$I_1 = \frac{V_1 - V_3}{2/S} = \frac{S}{2} V_1 - \frac{S}{2} V_3$$

$$\text{ie; } I_1 = \frac{S}{2} V_1 - \frac{S}{2} V_3 \quad \text{--- (C)}$$

Applying KCL at Node 3

$$\frac{V_1 - V_3}{2/s} = \frac{V_3}{s} + \frac{V_3 - V_2}{2/s}$$

$$\frac{s}{2} (V_1 - V_3) = \frac{V_3}{s} + \frac{s}{2} (V_3 - V_2)$$

$$\frac{s}{2} V_1 - \frac{s}{2} V_3 = \frac{V_3}{s} + \frac{s}{2} V_3 - \frac{s}{2} V_2$$

$$\frac{s}{2} V_1 + \frac{s}{2} V_2 = V_3 \left\{ \frac{1}{s} + \frac{s}{2} + \frac{s}{2} \right\}$$

$$\frac{s}{2} V_1 + \frac{s}{2} V_2 = V_3 \left\{ \frac{1}{s} + s \right\}$$

$$\frac{s}{2} V_1 + \frac{s}{2} V_2 = V_3 \left\{ \frac{s^2 + 1}{s} \right\}$$

$$V_3 = \frac{s^2}{2(s^2 + 1)} V_1 + \frac{s^2}{2(s^2 + 1)} V_2 \quad \text{--- (2)}$$

Substituting (2) in (1)

$$(1) \Rightarrow I_1 = \frac{s}{2} V_1 - \frac{s}{2} \left\{ \frac{s^2}{2(s^2 + 1)} V_1 + \frac{s^2}{2(s^2 + 1)} V_2 \right\}$$

$$I_1 = \frac{s}{2} V_1 - \frac{s^3}{4(s^2 + 1)} V_1 - \frac{s^3}{4(s^2 + 1)} V_2$$

$$I_1 = \frac{s^3 + 2s}{4(s^2 + 1)} V_1 - \frac{s^3}{4(s^2 + 1)} V_2 \quad \text{--- (A)}$$

Applying KCL at Node 2

$$I_2 = \frac{V_2}{S} + \frac{V_2 - V_3}{2/S}$$

$$I_2 = \frac{V_2}{S} + \frac{S}{2} V_2 - \frac{S}{2} V_3$$

$$= \left(\frac{1}{S} + \frac{S}{2} \right) V_2 - \frac{S}{2} V_3$$

$$I_2 = \frac{S^2 + 2}{2S} V_2 - \frac{S}{2} V_3 \quad \text{--- (3)}$$

Substituting (2) in (3)

$$I_2 = \frac{S^2 + 2}{2S} V_2 - \frac{S}{2} \left\{ \frac{S^2}{2(S^2 + 1)} V_1 + \frac{S^2}{2(S^2 + 1)} V_2 \right\}$$

$$= -\frac{S^3}{4(S^2 + 1)} V_1 + \left\{ \frac{S^2 + 2}{2S} - \frac{S^3}{4(S^2 + 1)} \right\} V_2$$

$$= -\frac{S^3}{4(S^2 + 1)} V_1 + \left\{ \frac{4S^4 + 8S^2 + 4S^2 + 8 - 2S^4}{8S(S^2 + 1)} \right\} V_2$$

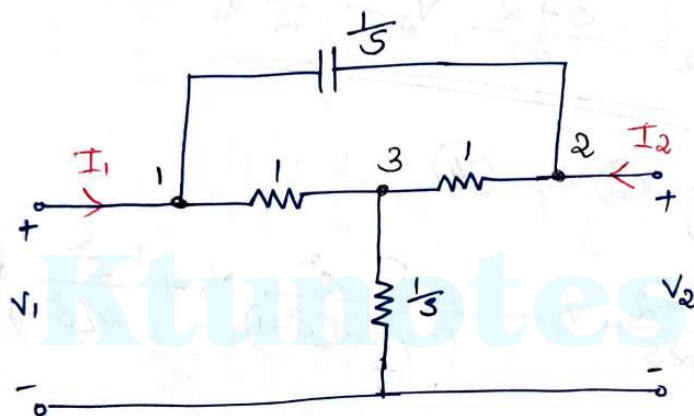
$$= -\frac{S^3}{4(S^2 + 1)} V_1 + \left\{ \frac{2S^4 + 12S^2 + 8}{8S(S^2 + 1)} \right\} V_2$$

$$I_2 = \frac{-S^3}{4(S^2 + 1)} V_1 + \left\{ \frac{S^4 + 6S^2 + 4}{4S(S^2 + 1)} \right\} V_2 \quad \text{--- (B)}$$

Comparing (A) & (B) with Y -parameter eqns.

$$\begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix} = \begin{bmatrix} \frac{s^3 + 2s}{4(s^2 + 1)} & \frac{-s^3}{4(s^2 + 1)} \\ \frac{-s^3}{4(s^2 + 1)} & \frac{s^4 + 6s^2 + 4}{4s(s^2 + 1)} \end{bmatrix}$$

6.



KCL at Node 1

$$I_1 = \frac{V_1 - V_3}{1} + \frac{V_1 - V_2}{1/s}$$

$$= V_1 - V_3 + sV_1 - sV_2$$

$$\underline{I_1 = (s+1)V_1 - sV_2 - V_3} \quad \text{--- (1)}$$

KCL at Node 2

$$I_2 = \frac{V_2 - V_3}{1} + \frac{V_2 - V_1}{1/s}$$

$$= V_2 - V_3 + sV_2 - sV_1$$

$$\underline{I_2 = -sV_1 + (s+1)V_2 - V_3} \quad \text{--- (2)}$$

KCL at Node 3

$$\frac{V_3 - V_1}{1} + \frac{V_3 - V_2}{1} + \frac{V_3}{1/s} = 0$$

$$V_3 - V_1 + V_3 - V_2 + sV_3 = 0$$

$$-V_1 - V_2 + (s+2)V_3 = 0$$

$$\underline{V_3 = \frac{1}{s+2} V_1 + \frac{1}{s+2} V_2} \quad \text{--- (3)}$$

③ in ①

$$\textcircled{1} \Rightarrow I_1 = (s+1)V_1 - sV_2 - \left\{ \frac{1}{s+2} V_1 + \frac{1}{s+2} V_2 \right\}$$

$$= \left\{ s+1 - \frac{1}{s+2} \right\} V_1 - \left\{ s + \frac{1}{s+2} \right\} V_2$$

$$= \left\{ \frac{(s+1)(s+2)-1}{s+2} \right\} V_1 - \left\{ \frac{s^2+2s+1}{s+2} \right\} V_2$$

$$\underline{I_1 = \left\{ \frac{s^2+3s+1}{s+2} \right\} V_1 - \left\{ \frac{s^2+2s+1}{s+2} \right\} V_2} \quad \text{--- (A)}$$

③ in ②

$$\textcircled{2} \Rightarrow I_2 = -sV_1 + (s+1)V_2 - \left\{ \frac{1}{(s+2)}V_1 + \frac{1}{(s+2)}V_2 \right\}$$

$$I_2 = -\left\{ s + \frac{1}{(s+2)} \right\} V_1 + \left\{ (s+1) - \frac{1}{(s+2)} \right\} V_2$$

$$I_2 = -\left\{ \frac{s^2 + 2s + 1}{s+2} \right\} V_1 + \left\{ \frac{s^2 + 3s + 1}{s+2} \right\} V_2 \quad \textcircled{B}$$

Comparing ① & ③ with γ -parameter eqns

$$\begin{bmatrix} \gamma_{11} & \gamma_{12} \\ \gamma_{21} & \gamma_{22} \end{bmatrix} = \begin{bmatrix} \frac{s^2 + 3s + 1}{s+2} & \frac{s^2 + 2s + 1}{s+2} \\ \frac{s^2 + 2s + 1}{s+2} & \frac{s^2 + 3s + 1}{s+2} \end{bmatrix}$$

TRANSMISSION PARAMETERS (ABCD PARAMETERS)

$$V_1 = A V_2 - B I_2$$

$$(V_1, I_1) = f(V_2, -I_2)$$

$$I_1 = C V_2 - D I_2$$

The current I_2 carries a negative sign. In transmission field, the output current is assumed to be coming out of the output port instead of going into the port.

$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} V_2 \\ -I_2 \end{bmatrix}$$

$$A = \left. \frac{V_1}{V_2} \right|_{I_2=0}$$

$I_2=0 \Rightarrow$ o/p port is open circuited

A is the reverse voltage gain with the o/p port open circuited.

$$C = \left. \frac{I_1}{V_2} \right|_{I_2=0}$$

C is the transfer admittance with the o/p port open circuited.

$$B = \left. -\frac{V_1}{I_2} \right|_{V_2=0}$$

$V_2=0 \Rightarrow$ o/p port is short circuited.

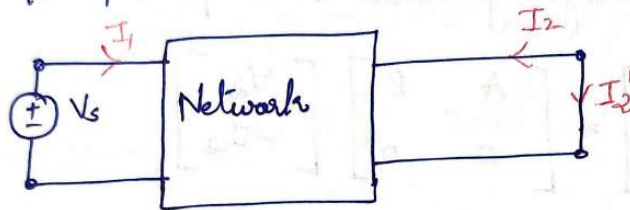
B is the transfer impedance with the o/p port short circuited.

$$D = - \frac{I_1}{I_2} \bigg|_{V_2=0}$$

D is the reverse current gain with the o/p port is short circuited.

Condition for Reciprocity.

a) Voltage V_s to i/p port.
o/p port short circuited.



$$V_1 = V_s, \quad V_2 = 0$$

$$I_2 = -I_2'$$

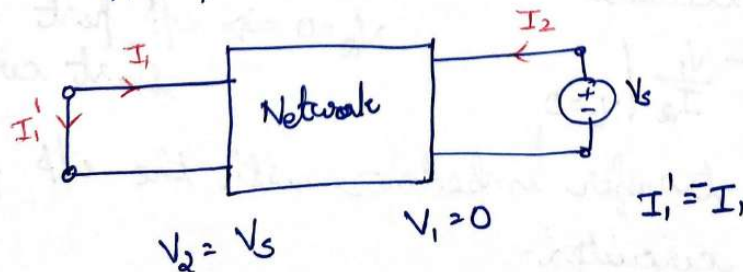
From transmission parameters

$$V_s = B I_2'$$

$$\underline{\underline{\frac{V_s}{I_2'} = B}}$$

$$\begin{aligned} V_1 &= A V_2 - B I_2 \\ I_1 &= C V_2 - D I_2 \end{aligned}$$

b) Voltage V_s to o/p port
i/p port short circuited



From the transmission parameter eqns

$$\begin{aligned} 0 &= AV_s - BI_2 \\ -I_1' &= CV_s - DI_2 \\ I_2 &= \frac{A}{B} V_s \\ -I_1' &= CV_s - \frac{AD}{B} V_s \end{aligned}$$

$$\frac{V_s}{I_1'} = -\frac{1}{C - \frac{AD}{B}} = \frac{B}{AD - BC}$$

$$\frac{V_s}{I_1'} = \frac{B}{AD - BC}$$

For the network to be reciprocal,

$$\frac{V_s}{I_2'} = \frac{V_s}{I_1'}$$

$$B = \frac{B}{AD - BC}$$

$$\underline{\underline{AD - BC = 1}}$$

Condition for Symmetry

a) When o/p port is open circuited; $I_2 = 0$

$$V_s = AV_2$$

$$I_1 = CV_2$$

$$\underline{\underline{\frac{V_s}{I_1} = \frac{A}{C}}}$$

b) When i/p port is open circuited; $I_1 = 0$

$$0 = CV_2 - DI_2$$

$$CV_2 = DI_2$$

$$\underline{\underline{\frac{V_s}{I_2} = \frac{D}{C}}}$$

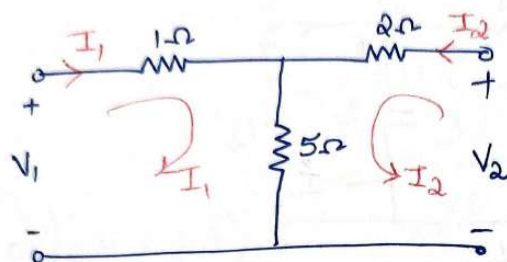
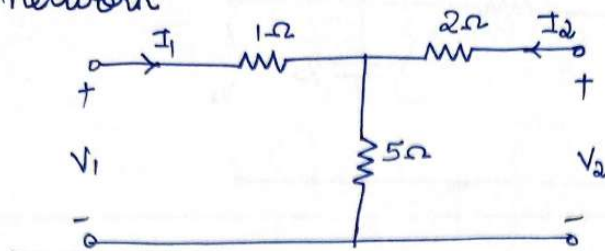
For the network to be symmetrical,

$$\frac{V_s}{I_1} = \frac{V_s}{I_2}$$

$$\frac{A}{C} = \frac{D}{C}$$

$$\underline{\underline{A = D}}$$

1. Find the transmission parameters for the network.



$$V_1 = A V_2 - B I_2$$

$$I_1 = C V_2 - D I_2$$

KVL to mesh 1 : $V_1 = I_1 + 5(I_1 + I_2)$

$$V_1 = 6I_1 + 5I_2 \quad \text{--- (1)}$$

KVL to mesh 2 : $V_2 = 2I_2 + 5(I_1 + I_2)$

$$V_2 = 5I_1 + 7I_2 \quad \text{--- (2)}$$

② $\Rightarrow 5I_1 = V_2 - 7I_2$

$$I_1 = \frac{1}{5}V_2 - \frac{7}{5}I_2 \quad \text{--- (A)}$$

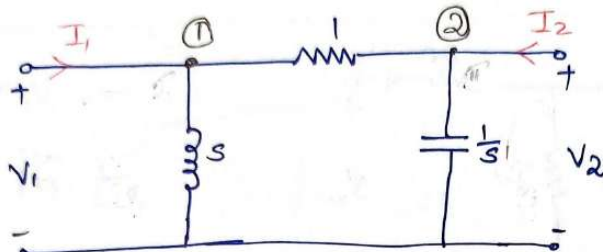
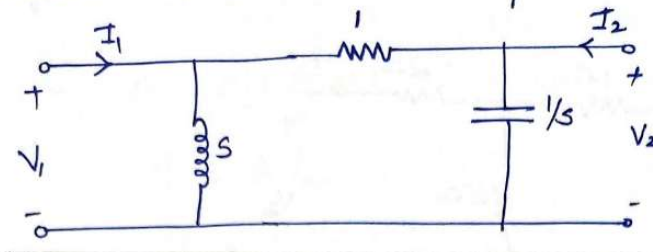
① $\Rightarrow V_1 = 6\left\{\frac{1}{5}V_2 - \frac{7}{5}I_2\right\} + 5I_2$

$$= \frac{6}{5}V_2 - \frac{17}{5}I_2 \quad \text{--- (B)}$$

Comparing (A) & (B) with transmission parameter eqns.

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} 6/5 & 17/5 \\ 1/5 & 7/5 \end{bmatrix}$$

2. Determine transmission parameters



KCL at ①

$$I_1 = \frac{V_1}{s} + \frac{V_1 - V_2}{1}$$

$$= V_1 \left\{ \frac{1}{s} + 1 \right\} - V_2 = \frac{(s+1)}{s} V_1 - V_2$$

$$I_1 = \frac{s+1}{s} V_1 - V_2 \quad \text{--- (c)}$$

KCL at ②

$$I_2 = \frac{V_2 - V_1}{1} + \frac{V_2}{1/s}$$

$$I_2 = V_2 - V_1 + sV_2$$

$$I_2 = -V_1 + (s+1)V_2$$

$$\therefore V_1 = -I_2 + (s+1)V_2$$

$$\therefore V_1 = (s+1)V_2 - I_2 \quad \text{--- (A)}$$

ⓐ

$$\textcircled{1} \Rightarrow I_1 = \frac{s+1}{s} V_1 - V_2$$

$$I_1 = \frac{s+1}{s} \{ (s+1)V_2 - I_2 \} - V_2$$

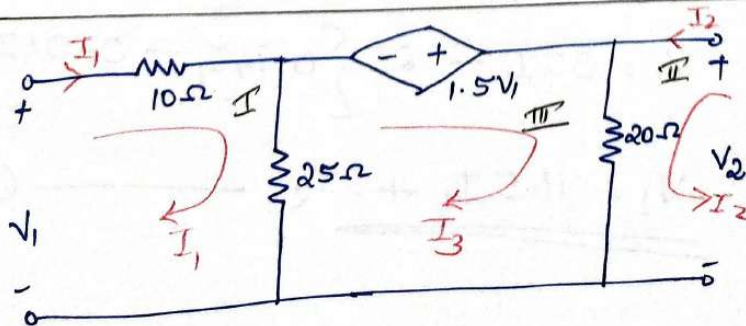
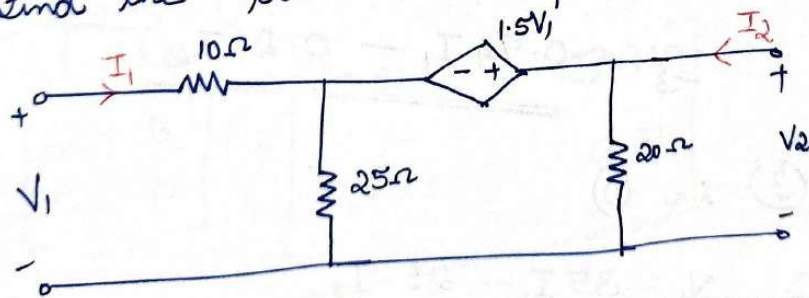
$$= \left\{ \frac{(s+1)^2}{s} - 1 \right\} V_2 - \frac{(s+1)}{s} I_2$$

$$= \left\{ \frac{s^2 + s + 1}{s} \right\} V_2 - \frac{(s+1)}{s} I_2 \quad \text{--- (B)}$$

Comparing $\textcircled{A}$ & $\textcircled{B}$ with ABCD parameter eqns

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} s+1 & 1 \\ \frac{s^2 + s + 1}{s} & \frac{s+1}{s} \end{bmatrix}$$

3. Find the transmission parameters.



KVL to Mesh 1

$$V_1 = 10I_1 + 25(I_1 - I_3)$$

$$V_1 = 35I_1 - 25I_3 \quad \text{————— (1)}$$

KVL to Mesh 2

$$V_2 = 20(I_2 + I_3)$$

$$V_2 = 20I_2 + 20I_3 \quad \text{————— (2)}$$

KVL to Mesh 3

$$-25(I_3 - I_1) + 1.5V_1 - 20(I_2 + I_3) = 0$$

$$25I_1 - 20I_2 - 45I_3 + 1.5V_1 = 0$$

$$25I_1 - 20I_2 - 45I_3 + 1.5\{35I_1 - 25I_3\} = 0$$

$$77.5I_1 - 20I_2 - 82.5I_3 = 0$$

$$\text{So } I_3 = \frac{77.5}{82.5}I_1 - \frac{20}{82.5}I_2$$

$$I_3 = \underline{\underline{0.94I_1 - 0.24I_2}} \quad \text{————— (3)}$$

Eqn (3) in (1)

$$\textcircled{1} \Rightarrow V_1 = 35I_1 - 25I_3$$

$$V_1 = 35I_1 - 25\{0.94I_1 - 0.24I_2\}$$

$$\underline{\underline{V_1 = 11.5I_1 + 6I_2}} \quad \text{————— (4)}$$

Eqn (3) in (2)

$$(2) \Rightarrow V_2 = 20I_2 + 20I_3$$

$$V_2 = 20I_2 + 20(0.94I_1 - 0.24I_2)$$

$$V_2 = 18.8I_1 + 15.2I_2 \quad \text{--- (5)}$$

$$\text{So } I_1 = \frac{V_2}{18.8} - \frac{15.2}{18.8}I_2$$

$$I_1 = 0.053V_2 - 0.81I_2 \quad \text{--- (A)}$$

Substituting (A) in (4)

$$V_1 = 11.5(0.053V_2 - 0.81I_2) + 6I_2$$

$$V_1 = 0.61V_2 - 3.32I_2 \quad \text{--- (B)}$$

Comparing (A) & (B) with ABCD parameters

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} 0.61 & 3.32 \\ 0.053 & 0.81 \end{bmatrix}$$

HYBRID PARAMETERS (h-PARAMETERS)

$$V_1 = h_{11} I_1 + h_{12} V_2$$

$$I_2 = h_{21} I_1 + h_{22} V_2$$

$$\begin{bmatrix} V_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ V_2 \end{bmatrix}$$

$$h_{11} = \left. \frac{V_1}{I_1} \right|_{V_2=0}$$

Short circuit impedance

$$h_{12} = \left. \frac{V_1}{V_2} \right|_{I_1=0}$$

Open circuit reverse voltage gain.

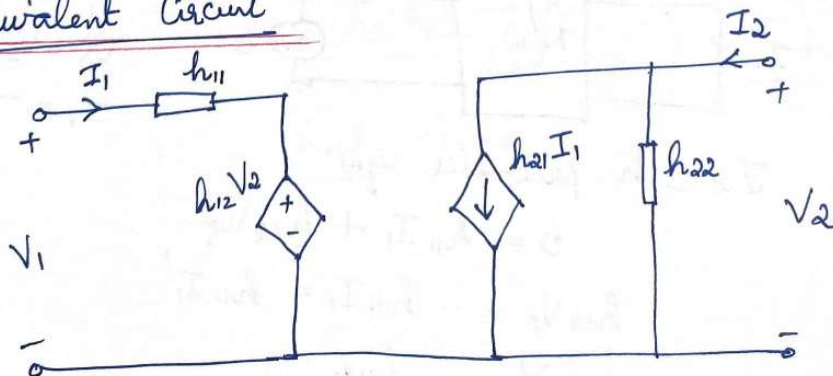
$$h_{21} = \left. \frac{I_2}{I_1} \right|_{V_2=0}$$

Short circuit forward current gain.

$$h_{22} = \left. \frac{I_2}{V_2} \right|_{I_1=0}$$

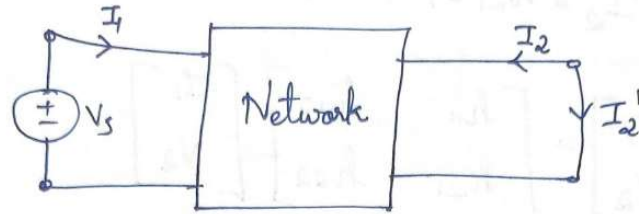
Open circuit output admittance.

Equivalent Circuit



Condition for Reciprocity

a) V_s is applied at i/p port & o/p port is short circuited



$$V_1 = V_s$$

$$V_2 = 0$$

$$I_2' = -I_2$$

$$V_1 = h_{11}I_1 + h_{12}I_2$$

$$I_2 = h_{21}I_1 + h_{22}I_2$$

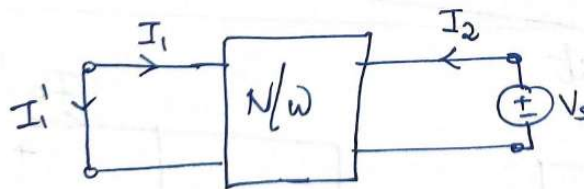
From h parameter eqns

$$V_s = h_{11}I_1$$

$$-I_2' = h_{21}I_1$$

$$\underline{\underline{\frac{V_s}{I_2'} = -\frac{h_{11}}{h_{21}}}}$$

b) V_s is applied at o/p port & i/p port is short circuited.



$$V_1 = 0$$

$$V_2 = V_s$$

$$I_1 = -I_1'$$

From h parameter eqns

$$0 = h_{11}I_1 + h_{12}V_s$$

$$h_{12}V_s = -h_{11}I_1 = h_{11}I_1'$$

$$\underline{\underline{\frac{V_s}{I_1'} = \frac{h_{11}}{h_{12}}}}$$

For network to be reciprocal,

$$\frac{V_s}{I_2} = \frac{V_s}{I_1}$$

$$\underline{h_{21} = -h_{12}}$$

Condition for Symmetry

The condition for symmetry is obtained from the z-parameters.

$$Z_{11} = \left. \frac{V_1}{I_1} \right|_{I_2=0} = \frac{h_{11} I_1 + h_{12} V_2}{I_1} \Big|_{I_2=0}$$

$$= \underline{h_{11} + h_{12} \frac{V_2}{I_1}}$$

But when $I_2=0$,

$$0 = h_{21} I_1 + h_{22} V_2$$

$$\frac{V_2}{I_1} = -\frac{h_{21}}{h_{22}}$$

$$Z_{11} = h_{11} - \frac{h_{12} h_{21}}{h_{22}}$$

$$= \frac{h_{11} h_{22} - h_{12} h_{21}}{h_{22}} = \underline{\underline{\frac{\Delta h}{h_{22}}}}$$

$$\underline{\underline{\Delta h = h_{11} h_{22} - h_{12} h_{21}}}$$

Similarly, $Z_{22} = \left. \frac{V_2}{I_2} \right|_{I_1=0}$

when $I_1=0$; $I_2 = h_{22} V_2$

So $Z_{22} = \left. \frac{V_2}{I_2} \right|_{I_1=0} = \frac{1}{h_{22}}$

For a symmetrical network,
 $Z_{11} = Z_{22}$

$$\frac{\Delta h}{h_{22}} = \frac{1}{h_{22}}$$

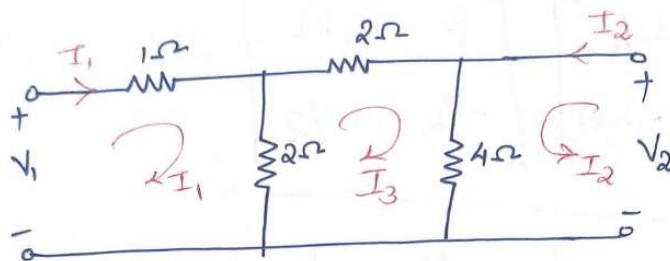
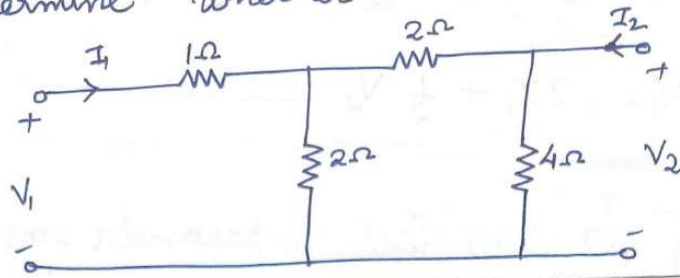
$$\Delta h = 1$$

ie; $h_{11} h_{22} - h_{12} h_{21} = 1$

Conditions for reciprocity and symmetry

Parameter	Condition for Reciprocity	Condition for symmetry
Z	$Z_{12} = Z_{21}$	$Z_{11} = Z_{22}$
Y	$Y_{12} = Y_{21}$	$Y_{11} = Y_{22}$
T	$AD - BC = 1$	$A = D$
h	$h_{12} = -h_{21}$	$h_{11} h_{22} - h_{12} h_{21} = 1$

1. Determine hybrid parameters for the network.
Determine whether the network is reciprocal.



$$V_1 = h_{11} I_1 + h_{12} V_2$$

$$I_2 = h_{21} I_1 + h_{22} V_2$$

$$\text{KVL to mesh 1; } V_1 = 3I_1 - 2I_3 \quad (1)$$

$$\text{KVL to mesh 2; } V_2 = 4I_2 + 4I_3 \quad (2)$$

$$\text{KVL to mesh 3; } -2(I_3 - I_1) - 2I_3 - 4(I_3 + I_2) = 0$$

$$2I_1 - 4I_2 - 8I_3 = 0$$

$$8I_3 = 2I_1 - 4I_2$$

$$I_3 = \frac{1}{4}I_1 - \frac{1}{2}I_2 \quad (3)$$

Substitute (3) in (1)

$$V_1 = 3I_1 - 2\left\{\frac{1}{4}I_1 - \frac{1}{2}I_2\right\} \quad (4)$$

$$\text{So } V_1 = \frac{5}{2}I_1 + I_2$$

Substitute (3) in (2)

$$V_2 = 4I_2 + 4\left\{\frac{1}{4}I_1 - \frac{1}{2}I_2\right\} \quad (5)$$

$$V_2 = I_1 + 2I_2$$

$$I_2 = \frac{1}{2}V_2 - \frac{1}{2}I_1 \quad (6)$$

Substituting (A) in (4)

$$V_1 = \frac{5}{2} I_1 + \frac{1}{2} V_2 - \frac{1}{2} I_1$$

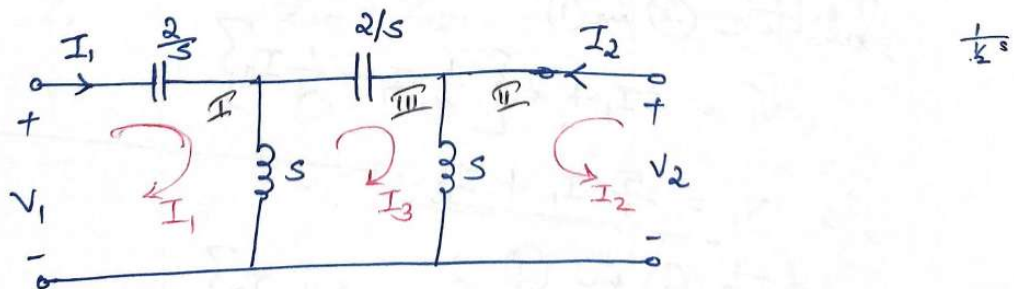
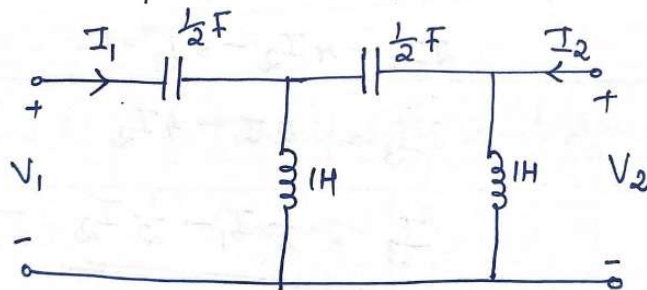
$$\underline{V_1 = 2 I_1 + \frac{1}{2} V_2} \quad \text{--- (B)}$$

Comparing (A) & (B) with h-parameter eqns

$$\begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} = \begin{bmatrix} 2 & 1/2 \\ -1/2 & 1/2 \end{bmatrix}$$

Since $\underline{h_{12} = -h_{21}}$, the network is reciprocal.

2. Find h-parameters for the network.



KVL to mesh 1; $V_1 = \left(\frac{2}{s} + s\right) I_1 - s I_3$

$$\underline{V_1 = \left(\frac{s+2}{s}\right) I_1 - s I_3} \quad \text{--- (1)}$$

KVL to mesh 2: $V_2 = 15(I_2 + I_3)$

$$V_2 = \underline{\underline{5I_2 + 5I_3}} \quad \text{--- (2)}$$

KVL to mesh 3: $-s(I_3 - I_1) - \frac{2}{s}I_3 - s(I_3 + I_2) = 0$
 $-sI_3 + sI_1 - \frac{2}{s}I_3 - sI_3 - sI_2 = 0$

So: $sI_1 - sI_2 - \left(2s + \frac{2}{s}\right)I_3 = 0$ $2s + \frac{2}{s}$

$$sI_1 - sI_2 - \left(\frac{2s^2 + 2}{s}\right)I_3 = 0$$

$$\left(\frac{2s^2 + 2}{s}\right)I_3 = sI_1 - sI_2$$

$$I_3 = \left(\frac{s^2}{2s^2 + 2}\right)I_1 - \left(\frac{s^2}{2s^2 + 2}\right)I_2 \quad \text{--- (3)}$$

Substituting (3) in (1)

$$\textcircled{1} \Rightarrow V_1 = \left\{\frac{s^2 + 2}{s}\right\}I_1 - \frac{s^3}{2s^2 + 2}I_1 + \frac{s^3}{2s^2 + 2}I_2$$

$$V_1 = \left\{\frac{s^2 + 2}{s} - \frac{s^3}{2s^2 + 2}\right\}I_1 + \frac{s^3}{2s^2 + 2}I_2$$

$$= \left\{\frac{2s^4 + 4s^2 + 2s^2 + 4 - s^4}{s(2s^2 + 2)}\right\}I_1 + \frac{s^3}{2s^2 + 2}I_2$$

$$V_1 = \underline{\underline{\frac{s^4 + 6s^2 + 4}{s(2s^2 + 2)}I_1 + \frac{s^3}{(2s^2 + 2)}I_2}} \quad \text{--- (4)}$$

Substitute (3) in (2)

$$(2) \Rightarrow V_2 = sI_2 + \frac{s^3}{2s^2+2} I_1 - \frac{s^3}{2s^2+2} I_2$$

$$= \left(\frac{s^3}{2s^2+2} \right) I_1 + \left\{ s - \frac{s^3}{(2s^2+2)} \right\} I_2$$

$$= \left\{ \frac{s^3}{2s^2+2} \right\} I_1 + \left\{ \frac{2s^3+2s-s^3}{2s^2+2} \right\} I_2$$

$$V_2 = \left\{ \frac{s^3}{2s^2+2} \right\} I_1 + \left\{ \frac{s^3+2s}{2s^2+2} \right\} I_2$$

$$I_2 = \left\{ \frac{2s^2+2}{s^3+2s} \right\} V_2 - \frac{s^3(2s^2+2)}{(2s^2+2)(s^3+2s)} I_1$$

$$I_2 = \left\{ \frac{2(s^2+1)}{s(s^2+2)} \right\} V_2 - \frac{s^2}{(s^2+2)} I_1 \quad \text{--- (A)}$$

$$I_2 = - \frac{s^2}{s^2+2} I_1 + \frac{2(s^2+1)}{s(s^2+2)} V_2 \quad \text{--- (A)}$$

Substituting (A) in (4)

$$(4) \Rightarrow V_1 = \frac{(s^4+6s^2+4)}{s(2s^2+2)} I_1 + \frac{s^3}{(2s^2+2)} \left\{ - \frac{s^2}{(s^2+2)} I_1 + \frac{2(s^2+1)}{s(s^2+2)} V_2 \right\}$$

$$V_1 = \left\{ \frac{(s^4+6s^2+4)(s^2+2)}{s(2s^2+2)(s^2+2)} - \frac{s^6}{s(2s^2+2)(s^2+2)} \right\} I_1 + \frac{2s^3(s^2+1)}{2(s^2+2)s(s^2+2)} V_2$$

$$V_1 = \frac{s^6 + 6s^4 + 4s^2 + 2s^4 + 12s^2 + 8 - s^6}{2s(s^2+1)(s^2+2)} I_1$$

$$+ \frac{s^2}{s^2+2} V_2$$

$$= \frac{8(s^4 + 2s^2 + 8)}{2s(s^2+1)(s^2+2)} I_1 + \frac{s^2}{s^2+2} V_2$$

$$= \frac{4(s^2+1)^2}{2s(s^2+1)(s^2+2)} I_1 + \frac{s^2}{s^2+2} V_2$$

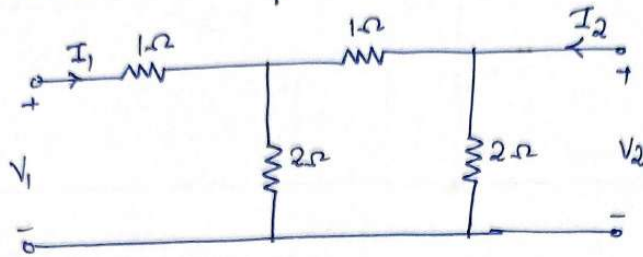
$$V_1 = \frac{4(s^2+1)}{s(s^2+2)} I_1 + \frac{s^2}{s^2+2} V_2 \quad \text{--- (3)}$$

Comparing (A) & (B) with h parameter eqns

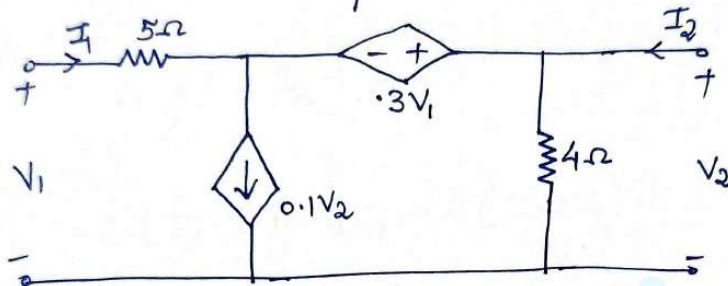
$$\begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} = \begin{bmatrix} \frac{4(s^2+1)}{s(s^2+2)} & \frac{s^2}{s^2+2} \\ -\frac{s^2}{s^2+2} & \frac{2(s^2+1)}{s(s^2+2)} \end{bmatrix}$$

Tutorial Questions

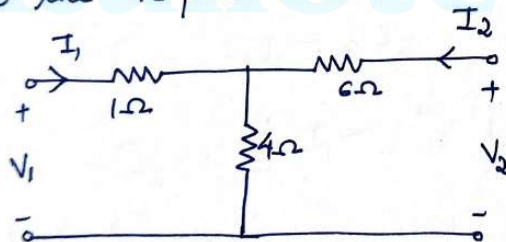
1. Obtain ABCD parameters



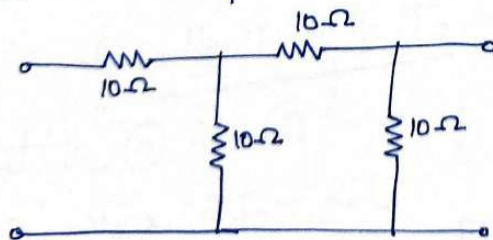
2. Find transmission parameters.



3. Find the h-parameters

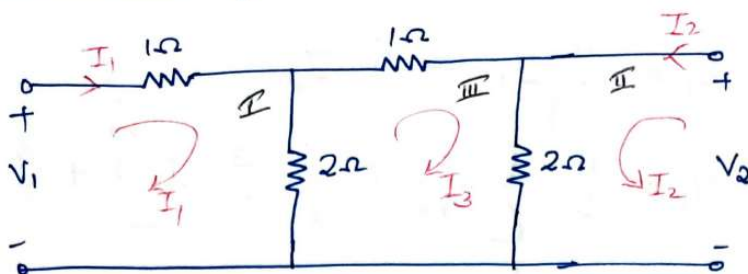


4. Find the h-parameters



Solutions for Tutorial

1.



KVL to Mesh 1

$$V_1 = 3I_1 - 2I_3 \quad (1)$$

KVL to mesh 2

$$V_2 = 2I_2 + 2I_3 \quad (2)$$

KVL to mesh 3

$$-2(I_3 - I_1) - 2I_3 - 2(I_2 + I_3) = 0$$

$$-2I_1 + 2I_2 + 5I_3 = 0 \quad (3)$$

$$5I_3 = 2I_1 - 2I_2$$

$$I_3 = \frac{2}{5}I_1 - \frac{2}{5}I_2 \quad (4)$$

Substitute (4) in (1)

$$(1) \Rightarrow V_1 = 3I_1 - 2 \left\{ \frac{2}{5}I_1 - \frac{2}{5}I_2 \right\} \quad (5)$$

$$V_1 = \frac{11}{5}I_1 + \frac{4}{5}I_2$$

Substitute (4) in (2)

$$(2) \Rightarrow V_2 = 2I_2 + 2 \left\{ \frac{2}{5}I_1 - \frac{2}{5}I_2 \right\}$$

$$V_2 = 2I_2 + \frac{4}{5}I_1 - \frac{4}{5}I_2$$

$$V_2 = \frac{4}{5}I_1 + \frac{6}{5}I_2$$

$$\text{So } I_1 = \frac{5}{4}V_2 - \frac{3}{2}I_2 \quad (A)$$

Substitute (A) in (5)

$$-\frac{33}{10} + \frac{8}{10} = -\frac{25}{10}$$

$$(5) \Rightarrow V_1 = \frac{11}{5} I_1 + \frac{4}{5} I_2$$

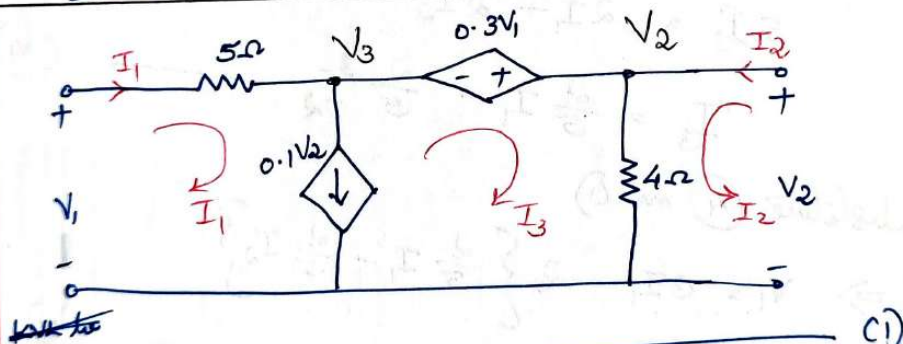
$$V_1 = \frac{11}{5} \left\{ \frac{5}{4} V_2 - \frac{3}{2} I_2 \right\} + \frac{4}{5} I_2$$

$$\underline{V_1 = \frac{11}{4} V_2 - \frac{5}{2} I_2} \quad \text{--- (B)}$$

Comparing (A) & (B) with ABCD parameter eqns

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} \frac{11}{4} & \frac{5}{2} \\ \frac{5}{4} & \frac{3}{2} \end{bmatrix}$$

2.



$$I_1 - I_3 = 0.1V_2 \quad \text{--- (1)}$$

KVL to mesh 1

$$V_1 - 5I_1 = V_3 \quad \text{--- (2)}$$

KVL to mesh 2

$$V_2 = 4(I_2 + I_3)$$

$$V_2 = 4I_2 + 4I_3$$

$$\text{So } I_3 = \frac{V_2}{4} - I_2 \quad \text{--- (3)}$$

Substitute ③ in ①

① $\Rightarrow$

$$I_1 - I_2 = 0.1V_2$$

$$I_1 - \left\{ \frac{V_2}{4} - I_2 \right\} = 0.1V_2$$

$$I_1 = \left(0.1 + \frac{1}{4}\right)V_2 - I_2$$

$$I_1 = 0.35V_2 - I_2 \quad \text{--- (A)}$$

KVL to mesh 3

$$V_3 + 0.3V_1 = V_2$$

$$\text{so } V_3 = V_2 - 0.3V_1 \quad \text{--- (4)}$$

Substitute ④ in ②

$$\text{② } \Rightarrow V_1 - 5I_1 = V_3$$

$$V_1 - 5I_1 = V_2 - 0.3V_1$$

$$1.3V_1 = V_2 + 5I_1$$

~~①~~ ~~give~~ Subst. value of I_1 from (A)

$$1.3V_1 = V_2 + 5 \left\{ 0.35V_2 - I_2 \right\}$$

$$= V_2 + 1.75V_2 - 5I_2$$

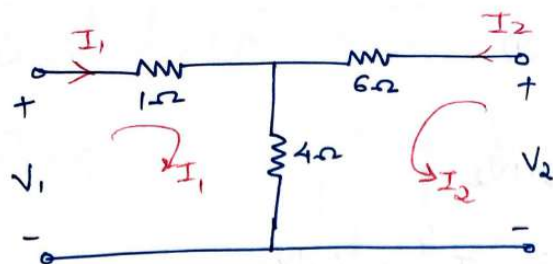
$$\text{ie; } 1.3V_1 = 2.75V_2 - 5I_2$$

$$\text{so } V_1 = 2.115V_2 - 3.846I_2 \quad \text{--- (B)}$$

from (A) & (B)

$$\text{so } \begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} 2.115 & 3.846 \\ 0.35 & 1 \end{bmatrix}$$

3.



$$\text{KVL to mesh 1; } V_1 = 5I_1 + 4I_2 \quad \text{--- (1)}$$

$$\text{KVL to mesh 2; } V_2 = 4I_1 + 10I_2 \quad \text{--- (2)}$$

$$\textcircled{2} \Rightarrow I_2 = \frac{V_2}{10} - \frac{4}{10} I_1$$

$$I_2 = -\frac{4}{10} I_1 + \frac{1}{10} V_2 \quad \text{--- (A)}$$

Substitute value of I_2 in eqn (1)

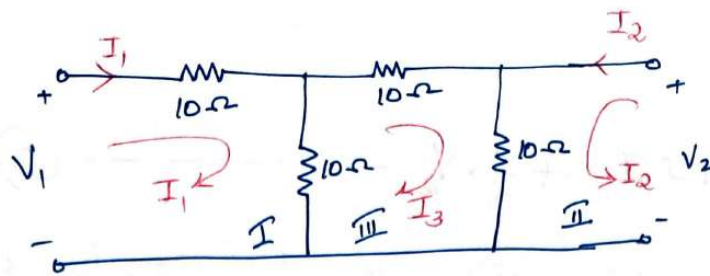
$$\textcircled{1} \Rightarrow V_1 = 5I_1 + 4 \left\{ -\frac{4}{10} I_1 + \frac{1}{10} V_2 \right\}$$

$$V_1 = \frac{34}{10} I_1 + \frac{4}{10} V_2 \quad \text{--- (B)}$$

Compare (A) & (B) with h parameter equations

$$\begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} = \begin{bmatrix} 3.4 & 0.4 \\ -0.4 & 0.1 \end{bmatrix}$$

4.



$$\text{KVL to mesh I} \rightarrow V_1 = 20I_1 - 10I_3 \quad \text{--- (1)}$$

$$\text{KVL to Mesh II} \rightarrow V_2 = 10I_2 + 10I_3 \quad \text{--- (2)}$$

$$\text{KVL to mesh III} \rightarrow 0 = -10I_1 + 10I_2 + 30I_3 \quad \text{--- (3)}$$

$$\textcircled{3} \Rightarrow I_3 = \frac{1}{3}I_1 - \frac{1}{3}I_2 \quad \text{--- (4)}$$

$$\textcircled{4} \text{ in } \textcircled{1} \Rightarrow V_1 = 20I_1 - 10 \left\{ \frac{1}{3}I_1 - \frac{1}{3}I_2 \right\}$$

$$\text{ie; } V_1 = \frac{50}{3}I_1 + \frac{10}{3}I_2 \quad \text{--- (5)}$$

$$\textcircled{5} \Rightarrow \textcircled{4} \text{ in } \textcircled{2}$$

$$\Rightarrow V_2 = 10I_2 + 10 \left\{ \frac{1}{3}I_1 - \frac{1}{3}I_2 \right\}$$

$$\text{ie; } V_2 = \frac{10}{3}I_1 + \frac{20}{3}I_2$$

$$\text{so } I_2 = \frac{3 \times 10}{20}I_1 + \frac{3V_2}{20}$$

$$I_2 = -\frac{1}{2}I_1 + \frac{3}{20}V_2 \quad \text{--- (A)}$$

Substitute value of I_2 in $\textcircled{5}$

$$\textcircled{5} \Rightarrow V_1 = \frac{50}{3}I_1 + \frac{10}{3} \left\{ -\frac{1}{2}I_1 + \frac{3}{20}V_2 \right\}$$

$$\text{ie; } V_1 = \left(\frac{50}{3} - \frac{5}{3} \right) I_1 + \frac{10 \times 3}{3 \times 20} V_2$$

~~$V_2 =$~~

$$V_1 = 15 I_1 + \frac{1}{2} V_2 \text{ ————— (B)}$$

Comparing (A) & (B) with h-parameter eqns

$$h_{11} = 15$$

$$h_{12} = \frac{1}{2}$$

$$h_{21} = -\frac{1}{2}$$

$$h_{22} = \frac{3}{20}$$

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INTER-RELATIONSHIP BETWEEN THE PARAMETERS

I

Z- PARAMETERS IN TERMS OF OTHER PARAMETERS

1. Z- Parameters in Terms of Y- parameters.

We know that

$$I_1 = Y_{11} V_1 + Y_{12} V_2$$

$$I_2 = Y_{21} V_1 + Y_{22} V_2$$

$$\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix}$$

$$V_1 = \frac{\begin{vmatrix} I_1 & Y_{12} \\ I_2 & Y_{22} \end{vmatrix}}{\begin{vmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{vmatrix}}} = \frac{Y_{22} I_1 - Y_{12} I_2}{Y_{11} Y_{22} - Y_{12} Y_{21}}$$

$$V_1 = \frac{Y_{22}}{\Delta Y} I_1 - \frac{Y_{12}}{\Delta Y} I_2 \quad \text{--- (A)}$$

$$\text{where } \Delta Y = Y_{11} Y_{22} - Y_{12} Y_{21}$$

Comparing (A) with

$$V_1 = Z_{11} I_1 + Z_{12} I_2$$

$$Z_{11} = \frac{Y_{22}}{\Delta Y}$$

$$Z_{12} = - \frac{Y_{12}}{\Delta Y}$$

$$\text{Also } V_2 = \frac{\begin{vmatrix} Y_{11} & I_1 \\ Y_{12} & I_2 \end{vmatrix}}{\begin{vmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{vmatrix}}} \\ = \frac{Y_{11} I_2 - Y_{12} I_1}{\Delta Y}$$

$$V_2 = -\frac{Y_{12}}{\Delta Y} I_1 + \frac{Y_{11}}{\Delta Y} I_2 \quad \text{--- (B)}$$

Comparing (B) with $V_2 = Z_{21} I_1 + Z_{22} I_2$

$$Z_{21} = -\frac{Y_{12}}{\Delta Y}$$

$$Z_{22} = \frac{Y_{11}}{\Delta Y}$$

2. Z-parameters in terms of ABCD parameters

we know that

$$V_1 = A V_2 + B I_2$$

$$I_1 = C V_2 + D I_2$$

$$V_2 = \frac{1}{C} I_1 + \frac{D}{C} I_2 \quad \text{--- (A)}$$

Comparing (A) with $V_2 = Z_{21} I_1 + Z_{22} I_2$

$$Z_{21} = \frac{1}{C}$$

$$Z_{22} = \frac{D}{C}$$

$$\text{Also } V_1 = A \left\{ \frac{1}{C} I_1 + \frac{D}{C} I_2 \right\} - B I_2$$

$$= \frac{A}{C} I_1 + \left(\frac{AD}{C} - B \right) I_2$$

$$V_1 = \underline{\underline{\frac{A}{C} I_1 + \frac{AD-BC}{C} I_2}} \quad \text{--- (B)}$$

Comparing (B) with $V_1 = Z_{11} I_1 + Z_{12} I_2$

$$Z_{11} = \frac{A}{C}$$

$$\underline{\underline{Z_{12} = \frac{AD-BC}{C}}}$$

3. Z- Parameters in terms of Hybrid parameters

We know that $V_1 = h_{11} I_1 + h_{12} V_2$

$$I_2 = h_{21} I_1 + h_{22} V_2$$

Rewriting the 2nd equation

$$V_2 = -\frac{h_{21}}{h_{22}} I_1 + \frac{1}{h_{22}} I_2 \quad \text{--- (A)}$$

Comparing (A) with $V_2 = Z_{21} I_1 + Z_{22} I_2$

$$Z_{21} = -\frac{h_{21}}{h_{22}}$$

$$\underline{\underline{Z_{22} = \frac{1}{h_{22}}}}$$

$$\text{Also } V_1 = h_{11} I_1 + h_{12} V_2$$

$$V_2 = h_{11} I_1 + h_{12} \left\{ -\frac{h_{21}}{h_{22}} I_1 + \frac{1}{h_{22}} I_2 \right\}$$

$$V_1 = \left\{ h_{11} - \frac{h_{12} \cdot h_{21}}{h_{22}} \right\} I_1 + \frac{h_{12}}{h_{22}} I_2$$

$$V_1 = \frac{h_{11} h_{22} - h_{12} h_{21}}{h_{22}} I_1 + \frac{h_{12}}{h_{22}} I_2 \quad (3)$$

Comparing (3) with $V_1 = Z_{11} I_1 + Z_{12} I_2$

$$Z_{11} = \frac{h_{11} h_{22} - h_{12} h_{21}}{h_{22}} = \frac{\Delta h}{h_{22}}$$

$$Z_{12} = \frac{h_{12}}{h_{22}}$$

II Y-Parameters in terms of Other Parameters

1. Y-Parameters in terms of Z-parameters

We know that $V_1 = Z_{11} I_1 + Z_{12} I_2$

$$V_2 = Z_{21} I_1 + Z_{22} I_2$$

$$I_1 = \frac{\begin{vmatrix} V_1 & Z_{12} \\ V_2 & Z_{22} \end{vmatrix}}{\begin{vmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{vmatrix}} = \frac{Z_{22} V_1 - Z_{12} V_2}{\Delta Z}$$

$$\text{where } \Delta Z = \underline{\underline{Z_{11} Z_{22} - Z_{12} Z_{21}}}$$

$$I_1 = \underline{\underline{\frac{Z_{22}}{\Delta Z} V_1 - \frac{Z_{12}}{\Delta Z} V_2}} \quad \text{--- (A)}$$

Comparing (A) with $I_1 = Y_{11} V_1 + Y_{12} V_2$

$$Y_{11} = \underline{\underline{\frac{Z_{22}}{\Delta Z}}}$$

$$Y_{12} = \underline{\underline{-\frac{Z_{12}}{\Delta Z}}}$$

$$\text{Also } I_2 = \frac{\begin{vmatrix} Z_{11} & V_1 \\ Z_{21} & V_2 \end{vmatrix}}{\begin{vmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{vmatrix}} = \frac{Z_{11} V_2 - Z_{21} V_1}{\Delta Z}$$

$$I_2 = \underline{\underline{-\frac{Z_{21}}{\Delta Z} V_1 + \frac{Z_{11}}{\Delta Z} V_2}} \quad \text{--- (B)}$$

Comparing (B) with $I_2 = Y_{21} V_1 + Y_{22} V_2$

$$Y_{21} = \underline{\underline{-\frac{Z_{21}}{\Delta Z}}}$$

$$Y_{22} = \underline{\underline{\frac{Z_{11}}{\Delta Z}}}$$

2. Y-Parameters in terms of ABCD Parameters

We know that $V_1 = A V_2 - B I_2$

$$I_1 = C V_2 - D I_2$$

Rewriting first eqn.

$$\underline{I_2 = -\frac{1}{B} V_1 + \frac{A}{B} V_2} \quad \text{--- (A)}$$

Comparing (A) with $I_2 = Y_{21} V_1 + Y_{22} V_2$

$$\underline{Y_{21} = -\frac{1}{B} ; \quad Y_{22} = \frac{A}{B}}$$

Also

$$I_1 = C V_2 - D I_2$$

$$= C V_2 - D \{ Y_{21} V_1 + Y_{22} V_2 \}$$

$$= C V_2 - D \left\{ -\frac{1}{B} V_1 + \frac{A}{B} V_2 \right\}$$

$$= \frac{D}{B} V_1 + \left(C - \frac{AD}{B} \right) V_2$$

$$\underline{I_1 = \frac{D}{B} V_1 + \left(\frac{AD - BC}{B} \right) V_2} \quad \text{--- (B)}$$

Comparing (B) with $I_1 = Y_{11} V_1 + Y_{12} V_2$

$$\underline{Y_{11} = \frac{D}{B} ; \quad Y_{12} = -\frac{(AD - BC)}{B} = -\frac{\Delta T}{B}}$$

3. Y-Parameters in terms of Hybrid Parameters

We know that

$$V_1 = h_{11} I_1 + h_{12} V_2$$

$$I_2 = h_{21} I_1 + h_{22} V_2$$

Rewriting the first eqn

$$I_1 = \frac{1}{h_{11}} V_1 - \frac{h_{12}}{h_{11}} V_2 \quad \text{--- (A)}$$

Comparing (A) with $I_1 = Y_{11} V_1 + Y_{12} V_2$

$$Y_{11} = \frac{1}{h_{11}} ; \quad Y_{12} = - \frac{h_{12}}{h_{11}}$$

Also $I_2 = h_{21} I_1 + h_{22} V_2$

$$= h_{21} \left\{ \frac{1}{h_{11}} V_1 - \frac{h_{12}}{h_{11}} V_2 \right\} + h_{22} V_2$$

$$= \frac{h_{21}}{h_{11}} V_1 + \left\{ h_{22} - \frac{h_{12} h_{21}}{h_{11}} \right\} V_2$$

$$I_2 = \frac{h_{21}}{h_{11}} V_1 + \frac{h_{11} h_{22} - h_{12} h_{21}}{h_{11}} V_2 \quad \text{--- (B)}$$

Comparing (B) with $I_2 = Y_{21} V_1 + Y_{22} V_2$

$$Y_{21} = \frac{h_{21}}{h_{11}}$$

$$Y_{22} = \frac{h_{11} h_{22} - h_{12} h_{21}}{h_{11}} = \frac{\Delta h}{h_{11}}$$

III

ABCD PARAMETERS IN TERMS OF OTHER PARAMETERS

1. ABCD Parameters in terms of Z-parameters

We know that

$$V_1 = Z_{11} I_1 + Z_{12} I_2$$

$$V_2 = Z_{21} I_1 + Z_{22} I_2$$

Rewriting the 2nd equation,

$$I_1 = \frac{1}{Z_{21}} V_2 - \frac{Z_{22}}{Z_{21}} I_2 \quad \text{--- (A)}$$

Comparing (A) with $I_1 = C V_2 - D I_2$,

$$C = \frac{1}{Z_{21}} \quad , \quad D = \frac{Z_{22}}{Z_{21}}$$

Also $V_1 = Z_{11} I_1 + Z_{12} I_2$

$$= Z_{11} \left\{ \frac{1}{Z_{21}} V_2 - \frac{Z_{22}}{Z_{21}} I_2 \right\} + Z_{12} I_2$$

$$= \frac{Z_{11}}{Z_{21}} V_2 - \left\{ \frac{Z_{11} Z_{22}}{Z_{21}} - Z_{12} \right\} I_2$$

$$V_1 = \frac{Z_{11}}{Z_{21}} V_2 - \left(\frac{Z_{11} Z_{22} - Z_{12} Z_{21}}{Z_{21}} \right) I_2 \quad \text{--- (B)}$$

Comparing (B) with $V_1 = A V_2 - B I_2$

$$A = \frac{Z_{11}}{Z_{21}} \quad ; \quad B = \frac{Z_{11} Z_{22} - Z_{12} Z_{21}}{Z_{21}} = \frac{\Delta Z}{Z_{21}}$$

2. ABCD Parameters in terms of Y-parameters

We know that

$$I_1 = Y_{11} V_1 + Y_{12} V_2$$

$$I_2 = Y_{21} V_1 + Y_{22} V_2$$

Rewriting the 2nd equation

$$V_1 = \frac{1}{Y_{21}} I_2 - \frac{Y_{22}}{Y_{21}} V_2$$

$$V_1 = -\frac{Y_{22}}{Y_{21}} V_2 + \frac{1}{Y_{21}} I_2 \quad \text{--- (A)}$$

Comparing (A) with $V_1 = A V_2 - B I_2$

$$A = \frac{Y_{22}}{Y_{21}} ; \quad B = \frac{1}{Y_{21}}$$

$$\text{Also } I_1 = Y_{11} \left\{ -\frac{Y_{22}}{Y_{21}} V_2 + \frac{1}{Y_{21}} I_2 \right\} + Y_{12} V_2$$

$$= \left\{ Y_{12} - \frac{Y_{11} Y_{22}}{Y_{21}} \right\} V_2 + \frac{Y_{11}}{Y_{21}} I_2$$

$$I_1 = \left\{ \frac{Y_{12} Y_{21} - Y_{11} Y_{22}}{Y_{21}} \right\} V_2 + \frac{Y_{11}}{Y_{21}} I_2 \quad \text{--- (B)}$$

Comparing (B) with $I_1 = C V_2 - D I_2$

$$C = \frac{Y_{12} Y_{21} - Y_{11} Y_{22}}{Y_{21}} = \frac{-\Delta Y}{Y_{21}}$$

$$D = \frac{Y_{11}}{Y_{21}}$$

3. ABCD Parameters in terms of Hybrid Parameters

We know that

$$V_1 = h_{11} I_1 + h_{12} V_2$$

$$I_2 = h_{21} I_1 + h_{22} V_2$$

Rewriting the 2nd eqn

$$I_1 = \frac{1}{h_{21}} I_2 - \frac{h_{22}}{h_{21}} V_2$$

$$I_1 = -\frac{h_{22}}{h_{21}} V_2 + \frac{1}{h_{21}} I_2 \quad \text{--- (A)}$$

Comparing (A) with $I_1 = C V_2 - D I_2$

$$C = -\frac{h_{22}}{h_{21}} ; \quad D = -\frac{1}{h_{21}}$$

~~$$\text{Also; } I_2 = h_{21} \left\{ -\frac{h_{22}}{h_{21}} V_2 + \frac{1}{h_{21}} I_2 \right\} + h_{22} V_2$$~~

~~$$= \left\{ h_{22} - \frac{h_{21} h_{22}}{h_{21}} \right\} V_2 + \frac{h_{21}}{h_{21}} I_2$$~~

$$\text{Also; } V_1 = h_{11} \left\{ -\frac{h_{22}}{h_{21}} V_2 + \frac{1}{h_{21}} I_2 \right\} + h_{12} V_2$$

$$V_1 = \left\{ h_{12} - \frac{h_{11} h_{22}}{h_{21}} \right\} V_2 + \frac{h_{11}}{h_{21}} I_2$$

$$V_1 = \left(\frac{h_{12} h_{21} - h_{11} h_{22}}{h_{21}} \right) V_2 + \frac{h_{11}}{h_{21}} I_2 \quad \text{--- (B)}$$

Comparing (B) with $V_1 = A V_2 - B I_2$

$$A = -\frac{\Delta h}{h_{21}}$$

$$B = -\frac{h_{11}}{h_{21}}$$

IV

HYBRID PARAMETERS IN TERMS OF OTHER PARAMETERS

1. Hybrid parameters in terms of Z-parameters.

We know that $V_1 = Z_{11} I_1 + Z_{12} I_2$

$$V_2 = Z_{21} I_1 + Z_{22} I_2$$

Reverting 2nd equation,

$$I_2 = \frac{1}{Z_{22}} V_2 - \frac{Z_{21}}{Z_{22}} I_1$$

$$I_2 = -\frac{Z_{21}}{Z_{22}} I_1 + \frac{1}{Z_{22}} V_2 \quad \text{--- (A)}$$

Comparing (A) with $I_2 = h_{21} I_1 + h_{22} V_2$

$$h_{21} = -\frac{Z_{21}}{Z_{22}} ; \quad h_{22} = \frac{1}{Z_{22}}$$

Also $V_1 = Z_{11} I_1 + Z_{12} \left\{ -\frac{Z_{21}}{Z_{22}} I_1 + \frac{1}{Z_{22}} V_2 \right\}$

$$V_1 = \left\{ Z_{11} - \frac{Z_{21} Z_{12}}{Z_{22}} \right\} I_1 + \frac{Z_{12}}{Z_{22}} V_2$$

$$V_1 = \left\{ \frac{Z_{11} \cdot Z_{22} - Z_{21} \cdot Z_{12}}{Z_{22}} \right\} I_1 + \frac{Z_{12}}{Z_{22}} V_2 \quad \text{--- (B)}$$

Comparing (B) with $V_1 = h_{11} I_1 + h_{12} V_2$

$$h_{11} = \frac{\Delta Z}{Z_{22}} ; \quad h_{12} = \frac{Z_{12}}{Z_{22}}$$

2. Hybrid Parameters in terms of Y-parameters

We know that $I_1 = Y_{11} V_1 + Y_{12} V_2$

$$I_2 = Y_{21} V_1 + Y_{22} V_2$$

Rewriting the first equation

$$V_1 = \frac{1}{Y_{11}} I_1 - \frac{Y_{12}}{Y_{11}} V_2 \quad (A)$$

Comparing (A) with $V_1 = h_{11} I_1 + h_{12} V_2$

$$\underline{h_{11} = \frac{1}{Y_{11}}} \quad ; \quad \underline{h_{12} = -\frac{Y_{12}}{Y_{11}}}$$

$$\text{Also } I_2 = Y_{21} \left\{ \frac{1}{Y_{11}} I_1 - \frac{Y_{12}}{Y_{11}} V_2 \right\} + Y_{22} V_2$$

$$= \frac{Y_{21}}{Y_{11}} I_1 - \left\{ \frac{Y_{21} Y_{12}}{Y_{11}} - Y_{22} \right\} V_2$$

$$= \frac{Y_{21}}{Y_{11}} I_1 + \left(\frac{Y_{11} Y_{22} - Y_{21} Y_{12}}{Y_{11}} \right) V_2 \quad (B)$$

Comparing (B) with $I_2 = h_{21} I_1 + h_{22} V_2$

$$\underline{h_{21} = \frac{Y_{21}}{Y_{11}}}$$

$$h_{22} = \frac{Y_{11} Y_{22} - Y_{12} Y_{21}}{Y_{11}}$$

$$\underline{h_{22} = \frac{\Delta Y}{Y_{11}}}$$

3. Hybrid Parameters in terms of ABCD Parameters

We know that

$$V_1 = AV_2 - BI_2$$

$$I_1 = CV_2 - DI_2$$

Rewriting 2nd eqn

$$I_2 = -\frac{1}{D} I_1 + \frac{C}{D} V_2 \quad \text{--- (A)}$$

Comparing (A) with $I_2 = h_{21} I_1 + h_{22} V_2$

$$h_{21} = -\frac{1}{D} ; \quad h_{22} = \frac{C}{D}$$

$$\text{Also } V_1 = AV_2 - B \left\{ -\frac{1}{D} I_1 + \frac{C}{D} V_2 \right\}$$

$$= \frac{B}{D} I_1 + \left(A - \frac{BC}{D} \right) V_2$$

$$V_1 = \frac{B}{D} I_1 + \frac{(AD - BC)}{D} V_2 \quad \text{--- (B)}$$

Comparing (B) with $V_1 = h_{11} I_1 + h_{12} V_2$

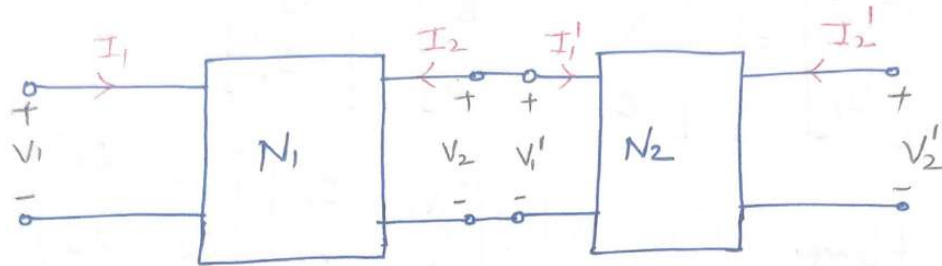
$$h_{11} = \frac{B}{D} ; \quad h_{12} = \frac{AD - BC}{D} = \frac{\Delta T}{D}$$

INTERCONNECTION OF TWO-PORT NETWORKS

I

Cascade Connection

Output Port of the first network becomes the input port of the second network.



$$I_1' = -I_2$$

Transmission Parameters of $N_1 \rightarrow A_1, B_1, C_1, D_1$

Transmission Parameters of $N_2 \rightarrow A_2, B_2, C_2, D_2$

For the N/w N_1

$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} A_1 & B_1 \\ C_1 & D_1 \end{bmatrix} \begin{bmatrix} V_2 \\ -I_2 \end{bmatrix} \quad \text{--- (1)}$$

For the N/w N_2

$$\begin{bmatrix} V_1' \\ I_1' \end{bmatrix} = \begin{bmatrix} A_2 & B_2 \\ C_2 & D_2 \end{bmatrix} \begin{bmatrix} V_2' \\ -I_2' \end{bmatrix} \quad \text{--- (2)}$$

Since $V_1' = V_2$ & $I_1' = -I_2$

$$\textcircled{2} \Rightarrow \begin{bmatrix} V_2 \\ -I_2 \end{bmatrix} = \begin{bmatrix} A_2 & B_2 \\ C_2 & D_2 \end{bmatrix} \begin{bmatrix} V_2' \\ -I_2' \end{bmatrix} \quad \text{--- (3)}$$

Combining (1) & (3)

$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} A_1 & B_1 \\ C_1 & D_1 \end{bmatrix} \begin{bmatrix} A_2 & B_2 \\ C_2 & D_2 \end{bmatrix} \begin{bmatrix} V_2' \\ -I_2' \end{bmatrix}$$

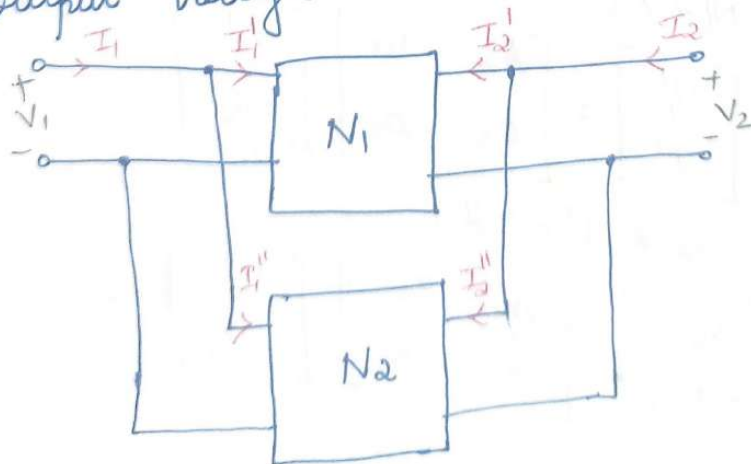
$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} V_2' \\ -I_2' \end{bmatrix}$$

$$\text{Hence; } \begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} A_1 & B_1 \\ C_1 & D_1 \end{bmatrix} \begin{bmatrix} A_2 & B_2 \\ C_2 & D_2 \end{bmatrix}$$

Resultant ABCD matrix of the cascade connection is the product of the individual ABCD matrices.

II Parallel Connection

In the parallel connection, the two networks have the same input voltages and the same output voltages.



Y-Parameters of N/w $N_1 \rightarrow Y_{11}', Y_{12}', Y_{21}', Y_{22}'$

Y-Parameters of N/w $N_2 \rightarrow Y_{11}'', Y_{12}'', Y_{21}'', Y_{22}''$

$$\text{For n/w } N_1, \begin{bmatrix} I_1' \\ I_2' \end{bmatrix} = \begin{bmatrix} Y_{11}' & Y_{12}' \\ Y_{21}' & Y_{22}' \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix}$$

For n/w N_2

$$\begin{bmatrix} I_1'' \\ I_2'' \end{bmatrix} = \begin{bmatrix} Y_{11}'' & Y_{12}'' \\ Y_{21}'' & Y_{22}'' \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix}$$

For the Combined network

$$I_1 = I_1' + I_1'' \quad \& \quad I_2 = I_2' + I_2''$$

$$\text{Hence } \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} I_1' + I_1'' \\ I_2' + I_2'' \end{bmatrix}$$

$$= \begin{bmatrix} Y_{11}' + Y_{11}'' & Y_{12}' + Y_{12}'' \\ Y_{21}' + Y_{21}'' & Y_{22}' + Y_{22}'' \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix}$$

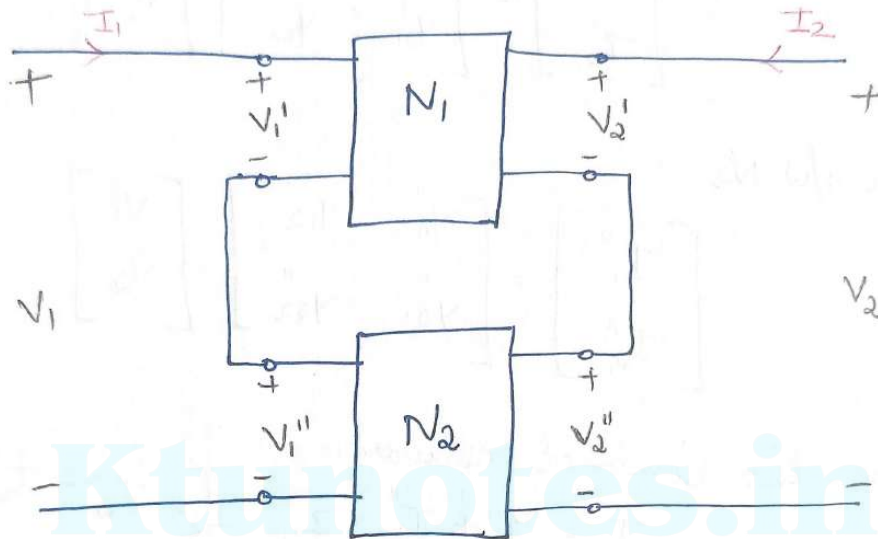
$$\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix}$$

Resultant Y matrix is the sum of Y matrices of each individual two port networks

III

Series Connection

In series Connection both the networks carry the same input Current. Their output currents are also equal.



Z parameters of $N_1 \rightarrow Z'_{11}, Z'_{12}, Z'_{21}, Z'_{22}$

Z parameters of $N_2 \rightarrow Z''_{11}, Z''_{12}, Z''_{21}, Z''_{22}$

For N_1

$$\begin{bmatrix} V_1' \\ V_2' \end{bmatrix} = \begin{bmatrix} Z'_{11} & Z'_{12} \\ Z'_{21} & Z'_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$$

For N_2

$$\begin{bmatrix} V_1'' \\ V_2'' \end{bmatrix} = \begin{bmatrix} Z''_{11} & Z''_{12} \\ Z''_{21} & Z''_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$$

For the Combined n/w $V_1 = V_1' + V_1''$ and $V_2 = V_2' + V_2''$

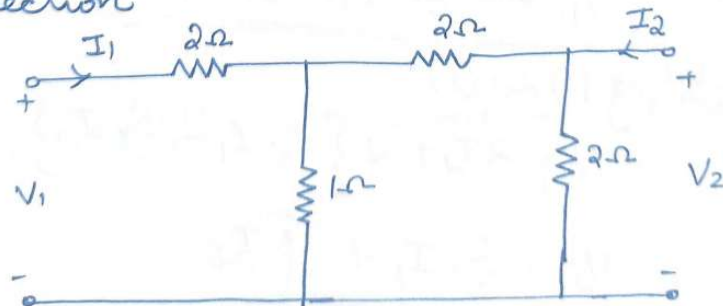
$$\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} V_1' + V_1'' \\ V_2' + V_2'' \end{bmatrix}$$

$$\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} Z_{11}' + Z_{11}'' & Z_{12}' + Z_{12}'' \\ Z_{21}' + Z_{21}'' & Z_{22}' + Z_{22}'' \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$$

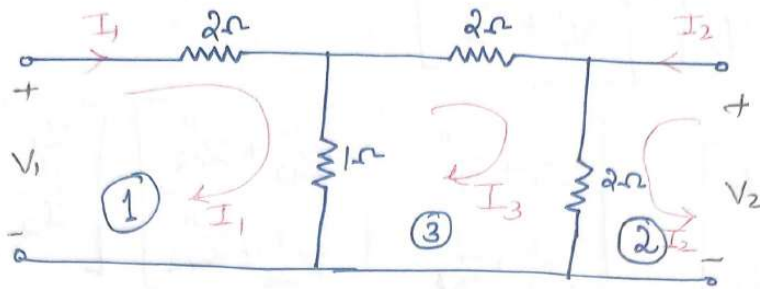
$$\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$$

Resultant Z-parameter matrix for the series connected network is the sum of Z-matrices of each individual 2 port network.

1. Two identical sections of the network shown in figure are connected in cascade. Obtain the transmission parameters of the overall connection



The network is redrawn



Applying KVL to Mesh 1

$$V_1 = 3I_1 - I_3 \quad \text{--- (1)}$$

Applying KVL to Mesh 2

$$V_2 = 2I_2 + 2I_3 \quad \text{--- (2)}$$

Applying KVL to Mesh 3

$$-I_1 + 2I_2 + 5I_3 = 0 \quad \text{--- (3)}$$

$$I_3 = \frac{1}{5}I_1 - \frac{2}{5}I_2 \quad \text{--- (3)}$$

Substituting (3) in (1)

$$V_1 = 3I_1 - \left\{ \frac{1}{5}I_1 - \frac{2}{5}I_2 \right\}$$

$$V_1 = \frac{14}{5}I_1 + \frac{2}{5}I_2 \quad \text{--- (4)}$$

Substituting (3) in (2)

$$V_2 = 2I_2 + 2 \left\{ \frac{1}{5}I_1 - \frac{2}{5}I_2 \right\}$$

$$V_2 = \frac{2}{5}I_1 + \frac{6}{5}I_2$$

$$I_1 = -3I_2 + \frac{5}{2}V_2 \quad \text{--- (A)}$$

Substituting (A) in (4)

$$V_1 = \frac{14}{5} \left\{ -3I_2 + \frac{5}{2}V_2 \right\} + \frac{2}{5}I_2$$

$$\underline{V_1 = -8I_2 + 7V_2} \quad \text{--- (B)}$$

Comparing (A) & B with ABCD parameter eqns
we get

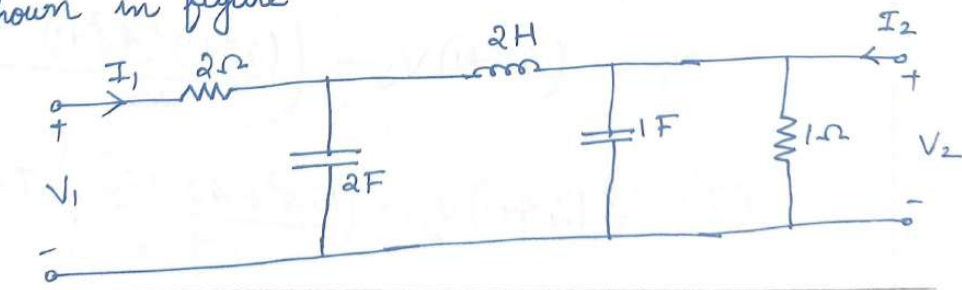
$$\begin{bmatrix} A_1 & B_1 \\ C_1 & D_1 \end{bmatrix} = \begin{bmatrix} 7 & +8 \\ 5/2 & +3 \end{bmatrix}$$

Hence transmission parameters of the overall N/w are

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} 7 & 8 \\ 5/2 & 3 \end{bmatrix} \begin{bmatrix} 7 & 8 \\ 5/2 & 3 \end{bmatrix}$$

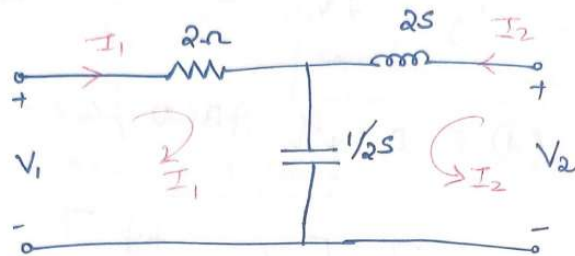
$$= \begin{bmatrix} 69 & 80 \\ 25 & 29 \end{bmatrix}$$

2. Determine ABCD parameters for the ladder N/w shown in figure.



The above N/w can be considered as a cascade connection of two networks N_1 & N_2 .

N/w N_1



Mesh 1 $\rightarrow V_1 = \left(2 + \frac{1}{2s}\right) I_1 + \frac{1}{2s} I_2$ ————— (1)

Mesh 2 $\rightarrow V_2 = \frac{1}{2s} I_1 + \left(2s + \frac{1}{2s}\right) I_2$ ————— (2)

From (2)

$$\Rightarrow I_1 = 2s V_2 - \left(2s + \frac{1}{2s}\right) 2s I_2$$

$$I_1 = 2s V_2 - (4s^2 + 1) I_2 \text{ ————— (A)}$$

(A) in (1)

$$(1) \Rightarrow V_1 = \left(2 + \frac{1}{2s}\right) \left\{ 2s V_2 - (4s^2 + 1) I_2 \right\} + \frac{1}{2s} I_2$$

$$V_1 = (4s + 1) V_2 - \left(\frac{(4s + 1)(4s^2 + 1)}{2s} - \frac{1}{2s} \right) I_2$$

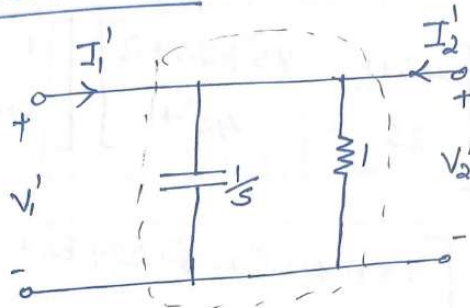
$$= (4s + 1) V_2 - \left(\frac{16s^3 + 4s + 4s^2 - 1}{2s} \right) I_2$$

$$V_1 = (4s + 1) V_2 - (8s^2 + 2s + 2) I_2 \text{ ————— (B)}$$

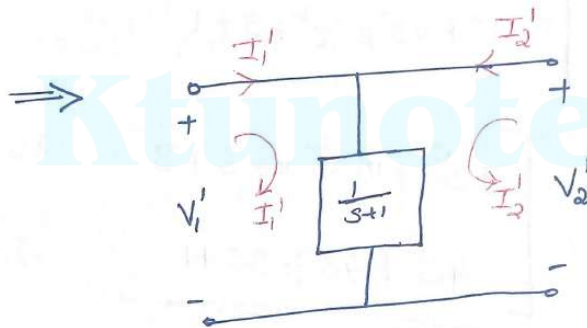
Comparing (A) & (B) with ABCD parameter eqns.

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} 4s+1 & 8s^2+2s+2 \\ 2s & 4s^2+1 \end{bmatrix}$$

Network N2



$$\left| \begin{array}{l} \frac{1}{s} \parallel 1 \\ = \frac{\frac{1}{s} \times 1}{\frac{1}{s} + 1} \\ = \frac{1}{s+1} \end{array} \right|$$



Mesh 1 $V_1' = \frac{1}{s+1} I_1' + \frac{1}{s+1} I_2'$ ————— (1)

Mesh 2 $V_2' = \frac{1}{s+1} I_1' + \frac{1}{s+1} I_2'$ ————— (2)

$I_1' = (s+1) V_2' - I_2'$ ————— (A)

① in ② $V_1' = \frac{1}{(s+1)} \left\{ (s+1) V_2' - I_2' \right\} + \frac{1}{(s+1)} I_2'$

$= V_2' - \frac{1}{(s+1)} I_2' + \frac{1}{s+1} I_2'$

$V_1' = V_2'$ ————— (B)

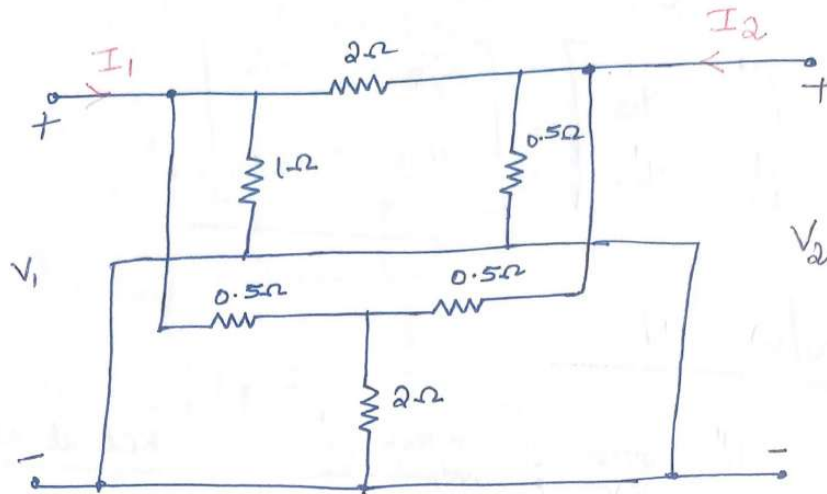
Comparing (A) & (B) with ABCD parameter eqns.

$$\begin{bmatrix} A_2 & B_2 \\ C_2 & D_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ s+1 & 1 \end{bmatrix}$$

Hence, overall ABCD parameters are

$$\begin{aligned} \begin{bmatrix} A & B \\ C & D \end{bmatrix} &= \begin{bmatrix} 4s+1 & 8s^2+2s+2 \\ 2s & 4s^2+1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ s+1 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 4s+1 + 8s^3+2s^2+2s+8s^2+2s+2 & 8s^2+2s+2 \\ 2s+4s^3+4s^2+s+1 & 4s^2+1 \end{bmatrix} \\ \begin{bmatrix} A & B \\ C & D \end{bmatrix} &= \begin{bmatrix} 8s^3+10s^2+8s+3 & 8s^2+2s+2 \\ 4s^3+4s^2+3s+1 & 4s^2+1 \end{bmatrix} \end{aligned}$$

3. Determine Y -parameters for the network shown



The given N/w can be considered as a parallel combination of 2 networks N_1 & N_2 .

N_1 N/w

KCL at Node 1

$$I_1' = \frac{V_1}{1} + \frac{V_1 - V_2}{2}$$

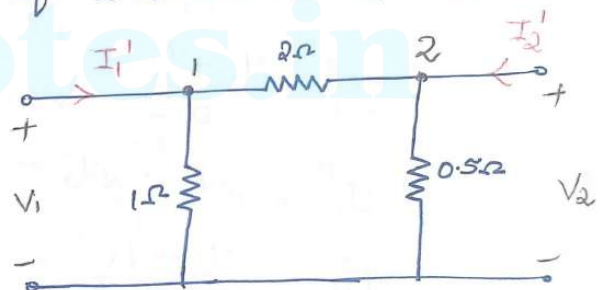
$$I_1' = \frac{3}{2} V_1 - \frac{1}{2} V_2 \quad (1)$$

KCL at Node 2

$$\frac{V_1 - V_2}{2} + I_2' = \frac{V_2}{0.5}$$

$$I_2' = \frac{-V_1 + V_2}{2} + \frac{V_2}{0.5}$$

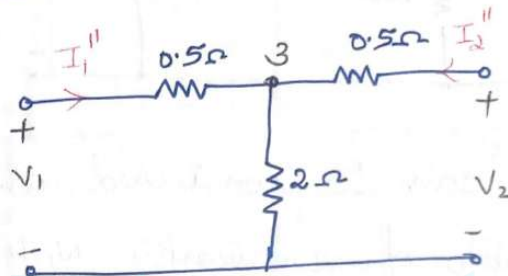
$$I_2' = -\frac{1}{2} V_1 + \frac{5}{2} V_2 \quad (2)$$



Comparing (1) & (2) with γ -parameters.

$$\begin{bmatrix} Y_{11}' & Y_{12}' \\ Y_{21}' & Y_{22}' \end{bmatrix} = \begin{bmatrix} 3/2 & -1/2 \\ -1/2 & 5/2 \end{bmatrix}$$

N/w N_2



KCL at Node 3

$$I_1'' + I_2'' = \frac{V_3}{2} \quad \text{--- (1)}$$

$$\frac{V_1 - V_3}{0.5} + \frac{V_2 - V_3}{0.5} = \frac{V_3}{2}$$

$$I_1'' = \frac{V_1 - V_3}{0.5}$$

$$I_1'' = 2V_1 - 2V_3 \quad \text{--- (2)}$$

$$I_2'' = \frac{V_2 - V_3}{0.5}$$

$$I_2'' = 2V_2 - 2V_3 \quad \text{--- (3)}$$

Substituting (2) & (3) ~~and~~ in (1).

$$2V_1 - 2V_3 + 2V_2 - 2V_3 = \frac{V_3}{2}$$

$$2V_1 + 2V_2 = \frac{9}{2} V_3$$

$$\therefore V_3 = \frac{4}{9} V_1 + \frac{4}{9} V_2 \quad \text{--- (4)}$$

Substituting (4) in (2)

$$\begin{aligned} (2) \Rightarrow I_1'' &= 2V_1 - 2 \left\{ \frac{4}{9} V_1 + \frac{4}{9} V_2 \right\} \\ I_1'' &= \frac{10}{9} V_1 - \frac{8}{9} V_2 \quad \text{--- (A)} \end{aligned}$$

Substituting (4) in (3)

$$\begin{aligned} (3) \Rightarrow I_2'' &= 2V_2 - 2 \left\{ \frac{4}{9} V_1 + \frac{4}{9} V_2 \right\} \\ I_2'' &= -\frac{8}{9} V_1 + \frac{10}{9} V_2 \quad \text{--- (B)} \end{aligned}$$

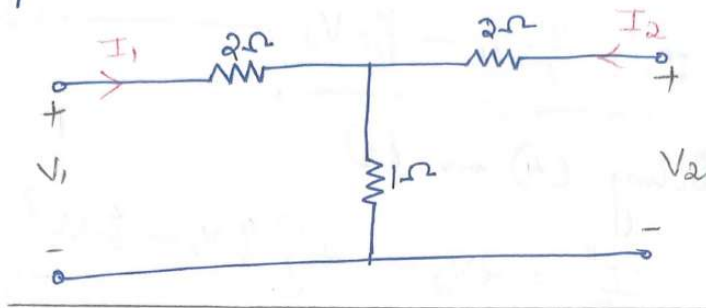
Comparing (A) & (B) with Y -parameter eqns

$$\begin{bmatrix} Y_{11}'' & Y_{12}'' \\ Y_{21}'' & Y_{22}'' \end{bmatrix} = \begin{bmatrix} \frac{10}{9} & -\frac{8}{9} \\ -\frac{8}{9} & \frac{10}{9} \end{bmatrix}$$

Hence Y -parameters of the overall network are

$$\begin{aligned} \begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix} &= \begin{bmatrix} Y_{11}' + Y_{11}'' & Y_{12}' + Y_{12}'' \\ Y_{21}' + Y_{21}'' & Y_{22}' + Y_{22}'' \end{bmatrix} \\ &= \begin{bmatrix} \frac{47}{18} & -\frac{25}{18} \\ -\frac{25}{18} & \frac{65}{18} \end{bmatrix} \end{aligned}$$

4. Two identical sections of the network shown in figure are connected in series. Obtain Z-parameters of the overall connection.



Applying KVL to mesh 1

$$V_1 = 3I_1 + I_2 \quad \text{--- (1)}$$

Applying KVL to mesh 2

$$V_2 = I_1 + 3I_2 \quad \text{--- (2)}$$

Comparing (1) & (2) with Z-parameter eqns

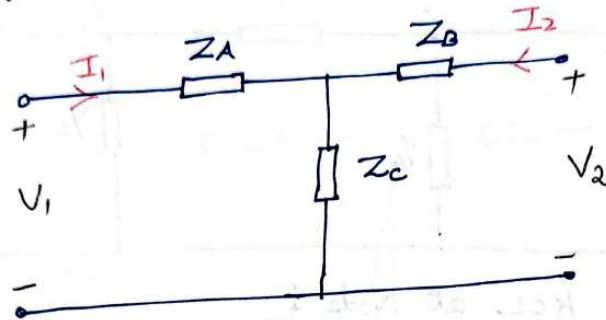
$$\begin{bmatrix} Z'_{11} & Z'_{12} \\ Z'_{21} & Z'_{22} \end{bmatrix} = \begin{bmatrix} Z''_{11} & Z''_{12} \\ Z''_{21} & Z''_{22} \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$$

Hence Z-parameters of the overall connection are

$$\begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} = \begin{bmatrix} 3+3 & 1+1 \\ 1+1 & 3+3 \end{bmatrix} = \begin{bmatrix} 6 & 2 \\ 2 & 6 \end{bmatrix}$$

T-NETWORK

Any two-port network can be represented by an equivalent T network as shown in figure.



Applying KVL to Mesh 1,

$$V_1 = (Z_A + Z_C) I_1 + Z_C I_2 \quad \text{--- (1)}$$

Applying KVL to Mesh 2,

$$V_2 = Z_C I_1 + (Z_B + Z_C) I_2 \quad \text{--- (2)}$$

Comparing (1) & (2) with Z-parameter eqns

$$Z_{11} = Z_A + Z_C$$

$$Z_{12} = Z_C$$

$$Z_{21} = Z_C$$

$$Z_{22} = Z_B + Z_C$$

Solving the above eqns

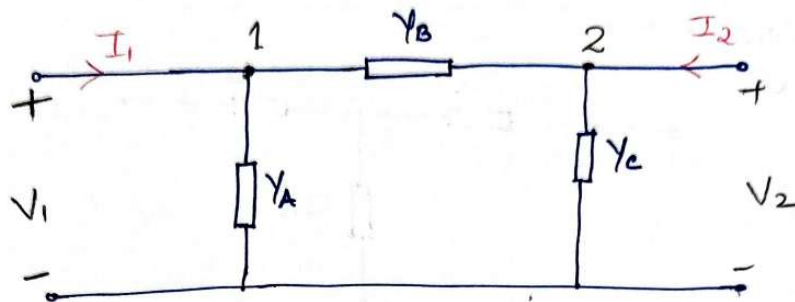
$$Z_A = Z_{11} - Z_{12}$$

$$Z_B = Z_{22} - Z_{12}$$

$$Z_C = Z_{12} = Z_{21}$$

PI (π) - NETWORK

Any two-port network can be represented by an equivalent π (π)



Applying KCL at Node 1

$$I_1 = Y_A V_1 + Y_B (V_1 - V_2)$$

$$I_1 = (Y_A + Y_B) V_1 - Y_B V_2 \quad \text{--- (1)}$$

Applying KCL at Node 2

$$I_2 = Y_C V_2 - Y_B (V_1 - V_2)$$

$$I_2 = -Y_B V_1 + (Y_B + Y_C) V_2 \quad \text{--- (2)}$$

Comparing (1) & (2) with Y -parameter eqns.

$$Y_{11} = Y_A + Y_B$$

$$Y_{22} = Y_B + Y_C$$

$$Y_{12} = -Y_B$$

$$Y_{21} = -Y_B$$

Solving above eqns

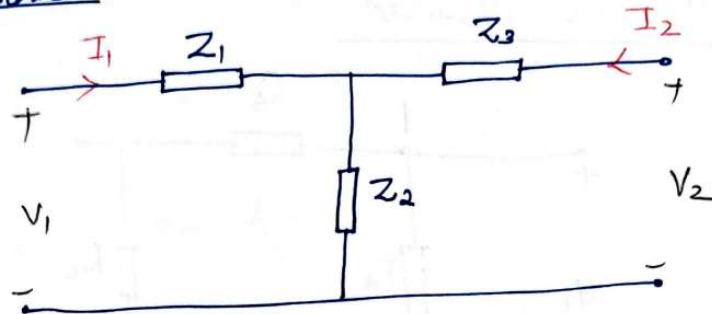
$$Y_A = Y_{11} + Y_{12} = Y_{11} + Y_{21}$$

$$Y_B = -Y_{12}$$

$$Y_C = Y_{22} + Y_{12} = Y_{22} + Y_{21}$$

1. The Z-parameters of a two port network are: $Z_{11} = 10\Omega$, $Z_{12} = Z_{21} = 5\Omega$, $Z_{22} = 20\Omega$. Find the equivalent T. network

T. Network



Applying KVL to Mesh 1

$$V_1 = (Z_1 + Z_2) I_1 + Z_2 I_2 \quad \text{--- (1)}$$

Applying KVL to Mesh 2

$$V_2 = Z_2 I_1 + (Z_2 + Z_3) I_2 \quad \text{--- (2)}$$

Comparing ① & ② with Z-parameter eqns

$$Z_{11} = Z_1 + Z_2 = 10$$

$$Z_{12} = Z_2 = 5$$

$$Z_{21} = Z_2 = 5$$

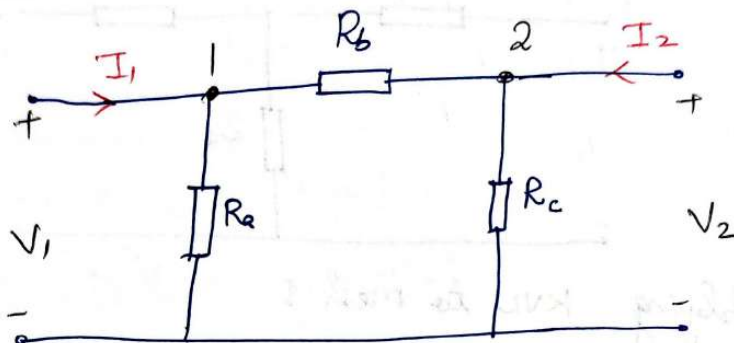
$$Z_{22} = Z_2 + Z_3 = 20\Omega$$

Solving the above eqns

$$\underline{\underline{Z_1 = 5\Omega, \quad Z_2 = 5\Omega, \quad Z_3 = 15\Omega}}$$

2. Admittance parameters of a γ -network are $Y_{11} = 0.09 \text{ S}$, $Y_{12} = Y_{21} = -0.05 \text{ S}$ and $Y_{22} = 0.07 \text{ S}$. Find the values of R_a , R_b and R_c .

The Pi Network



Applying KCL at Node 1

$$I_1 = \frac{V_1}{R_a} + \frac{V_1 - V_2}{R_b}$$

$$I_1 = \left[\frac{1}{R_a} + \frac{1}{R_b} \right] V_1 - \frac{1}{R_b} V_2 \quad \text{--- (1)}$$

Applying KCL at Node 2

$$I_2 + \frac{V_1 - V_2}{R_b} = \frac{V_2}{R_c}$$

$$I_2 = -\frac{1}{R_b} V_1 + \left(\frac{1}{R_b} + \frac{1}{R_c} \right) V_2 \quad \text{--- (2)}$$

Comparing (1) & (2) with γ -parameter eqns

$$Y_{11} = \frac{1}{R_a} + \frac{1}{R_b} = 0.09$$

$$Y_{12} = -\frac{1}{R_b} = -0.05$$

$$Y_{21} = -\frac{1}{R_b} = 0.05$$

$$Y_{22} = \frac{1}{R_b} + \frac{1}{R_c} = 0.07$$

Solving the above eqns

$$R_b = \frac{1}{0.05} = \underline{\underline{20\Omega}}$$

$$\begin{aligned} \frac{1}{R_a} &= 0.09 - \frac{1}{R_b} \\ &= 0.09 - 0.05 = 0.04 \end{aligned}$$

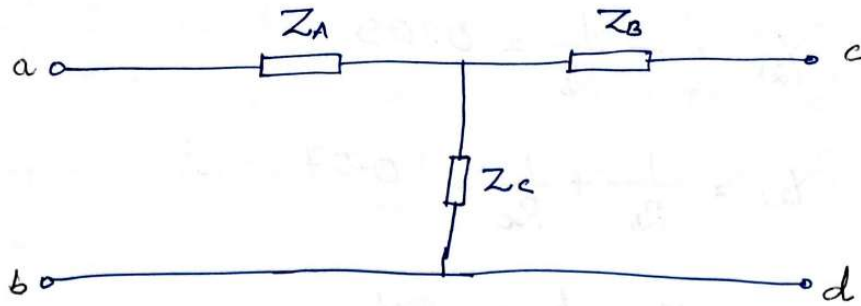
$$R_a = \frac{1}{0.04} = \underline{\underline{25\Omega}}$$

$$\frac{1}{R_c} = 0.07 - \frac{1}{R_b} = 0.07 - 0.05$$

$$\frac{1}{R_c} = 0.02; \quad R_c = \frac{1}{0.02} = \underline{\underline{50\Omega}}$$

$$\underline{\underline{\text{So } R_a = 25\Omega, R_b = 20\Omega \text{ \& } R_c = 50\Omega}}.$$

3. A network has two input terminals (a, b) and two output terminals (c, d) as shown in fig. The input impedance with c & d open circuited is $(250 + j100)\Omega$ and with c & d short circuited is $(400 + j300)\Omega$. The impedance across c & d with a & b open circuited is 200Ω . Determine the T. network parameters.



Input impedance with c & d open circuited is

$$(Z_A + Z_C) = (250 + j100) \quad \text{--- (1)}$$

Input impedance with c & d short circuited is

$$Z_A + \frac{Z_B Z_C}{Z_B + Z_C} = 400 + j300 \quad \text{--- (2)}$$

Impedance across c & d with a & b Open Circuited

$$Z_B + Z_C = 200 \quad \text{--- (3)}$$

② - ①

$$\Rightarrow \frac{Z_B Z_C}{Z_B + Z_C} - Z_C = 150 + j200 \quad \text{--- (4)}$$

$$\text{From (3)} \Rightarrow Z_B = 200 - Z_C$$

substitute Z_B in (4)

$$\text{④} \Rightarrow \frac{(200 - Z_C) Z_C}{200 - Z_C + Z_C} - Z_C = 150 + j200$$

$$\frac{200 Z_C - Z_C^2}{200} - Z_C = 150 + j200$$

$$200Z_c - Z_c^2 - 200Z_c = 200(150 + j200)$$

$$-Z_c^2 = 200(150 + j200)$$

$$-Z_c^2 = 30000 + j40000$$

$$Z_c^2 = -30000 - j40000$$

$$Z_c = (100 - j200)^2$$

$$\underline{\underline{Z_c = 100 - j200}}$$

$$Z_B = 200 - Z_c = \underline{\underline{(100 + j200) \Omega}}$$

$$Z_A = 250 + j100 - Z_c$$

$$= 250 + j100 - 100 + j200$$

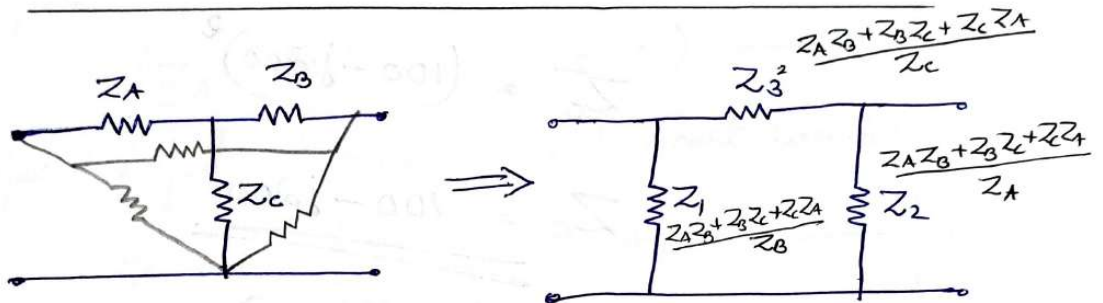
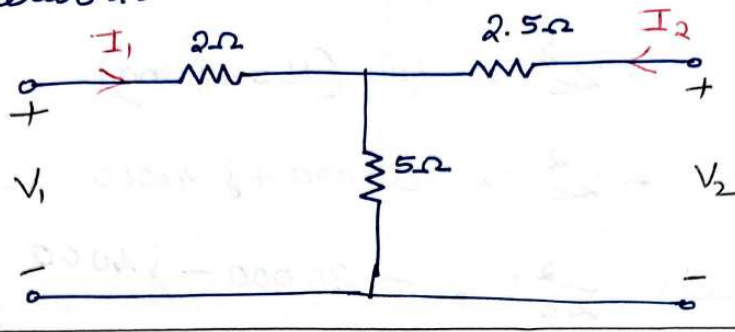
$$\underline{\underline{Z_A = (150 + j300) \Omega}}$$

$$\text{So } Z_A = (150 + j300) \Omega$$

$$Z_B = (100 + j200) \Omega$$

$$\underline{\underline{Z_c = (100 - j200) \Omega}}$$

4. Find the equivalent π network for the T network shown



T (star) to Equivalent π (Delta)

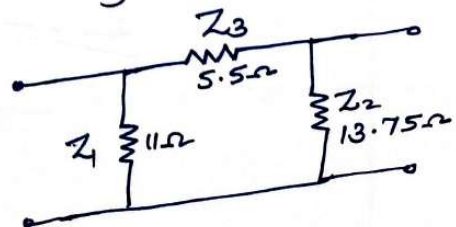
$$Z_1 = \frac{Z_A Z_B + Z_B Z_C + Z_C Z_A}{Z_B}$$

$$= \frac{2 \times 2.5 + 2.5 \times 5 + 5 \times 2}{2.5} = \frac{5 + 12.5 + 10}{2.5}$$

$$= \frac{27.5}{2.5} = \underline{\underline{11\Omega}}$$

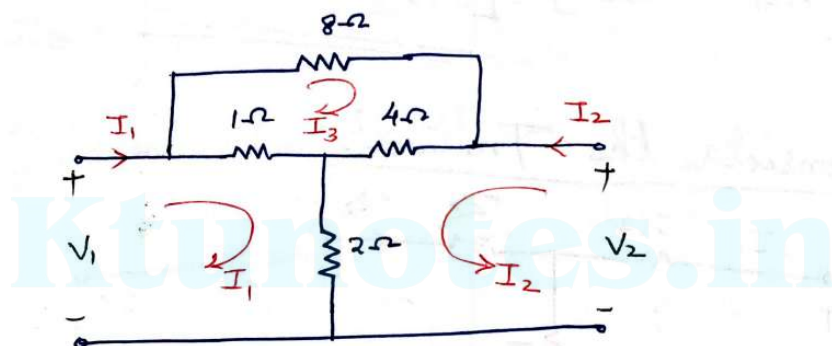
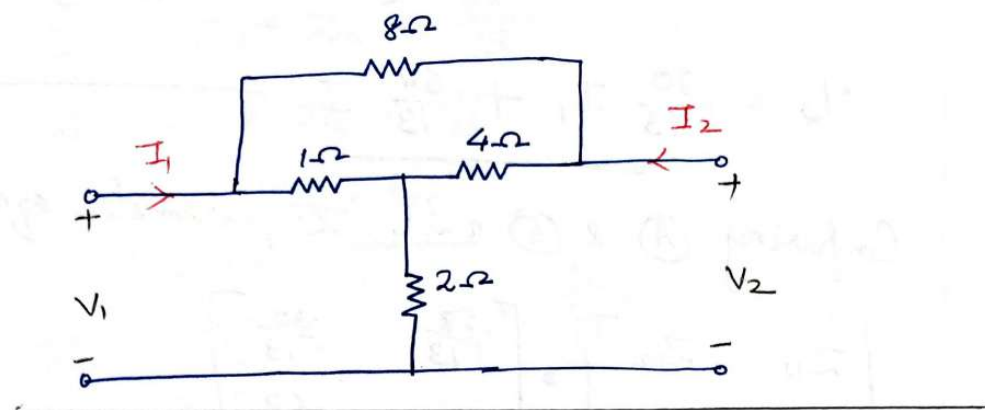
$$Z_2 = \frac{Z_A Z_B + Z_B Z_C + Z_C Z_A}{Z_A} = \frac{27.5}{2} = \underline{\underline{13.75\Omega}}$$

$$Z_3 = \frac{Z_A Z_B + Z_B Z_C + Z_C Z_A}{Z_C} = \frac{27.5}{5} = \underline{\underline{5.5\Omega}}$$



5

For the network shown, find equivalent T Network



KVL to Mesh 1

$$V_1 = 3I_1 + 2I_2 - I_3 \quad (1)$$

KVL to Mesh 2

$$V_2 = 2I_1 + 6I_2 + 4I_3 \quad (2)$$

KVL to Mesh 3

$$0 = -I_1 + 4I_2 + 13I_3$$

$$I_3 = \frac{1}{13} I_1 - \frac{4}{13} I_2 \quad (3)$$

Substitute (3) in (1)

$$V_1 = 3I_1 + 2I_2 - \left\{ \frac{1}{13} I_1 - \frac{4}{13} I_2 \right\}$$

$$V_1 = \frac{38}{13} I_1 + \frac{30}{13} I_2 \quad (4)$$

Substitute (3) in (2)

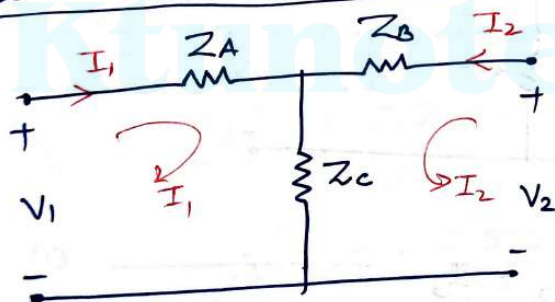
$$V_2 = 2I_1 + 6I_2 + 4 \left\{ \frac{1}{13} I_1 - \frac{4}{13} I_2 \right\}$$

$$V_2 = \frac{30}{13} I_1 + \frac{62}{13} I_2 \quad \text{--- (B)}$$

Comparing (A) & (B) with Z-parameter eqns

$$\begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} = \begin{bmatrix} \frac{38}{13} & \frac{30}{13} \\ \frac{30}{13} & \frac{62}{13} \end{bmatrix}$$

Consider the T network



KVL to Mesh 1

$$V_1 = (Z_A + Z_C) I_1 + Z_C I_2$$

KVL to Mesh 2

$$V_2 = Z_C I_1 + (Z_B + Z_C) I_2$$

∴ Z parameters are

$$\begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} = \begin{bmatrix} Z_A + Z_C & Z_C \\ Z_C & Z_B + Z_C \end{bmatrix}$$

Comparing Z parameters

$$Z_{11} = Z_A + Z_C = \frac{38}{13}$$

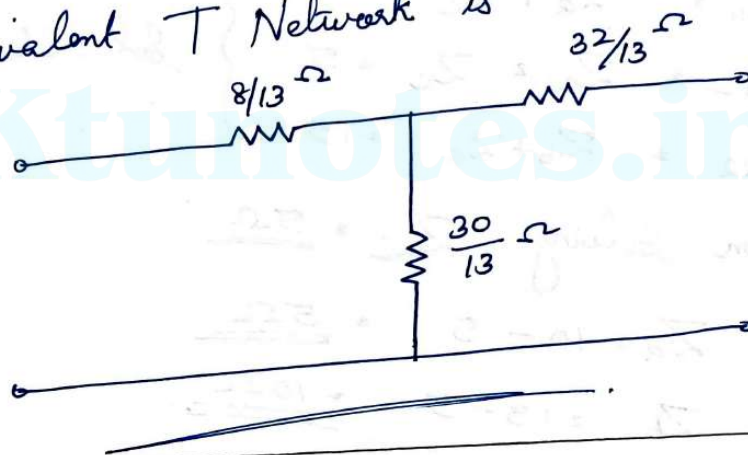
$$Z_{12} = Z_{21} = Z_C = \frac{30}{13}$$

$$Z_{22} = Z_B + Z_C = \frac{62}{13}$$

$$Z_C = \frac{30}{13}; \quad Z_A = \frac{38}{13} - \frac{30}{13} = \frac{8}{13}$$

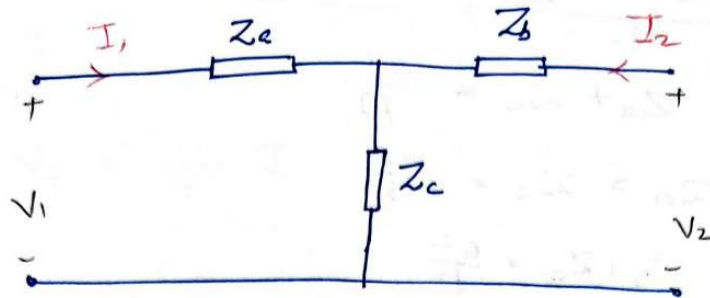
$$Z_B = \frac{62}{13} - \frac{30}{13} = \frac{32}{13}$$

Equivalent T Network is



6. The Z parameters of a two port network are $Z_{11} = 10\Omega$, $Z_{22} = 15\Omega$, $Z_{12} = Z_{21} = 5\Omega$. Find the equivalent T network and ABCD parameters.

T- Network



$$V_1 = (Z_a + Z_c) I_1 + Z_c I_2$$

$$V_2 = Z_c I_1 + (Z_b + Z_c) I_2$$

Comparing with Z parameter eqns

$$Z_{11} = Z_a + Z_c = 10$$

$$Z_{12} = Z_{21} = Z_c = 5$$

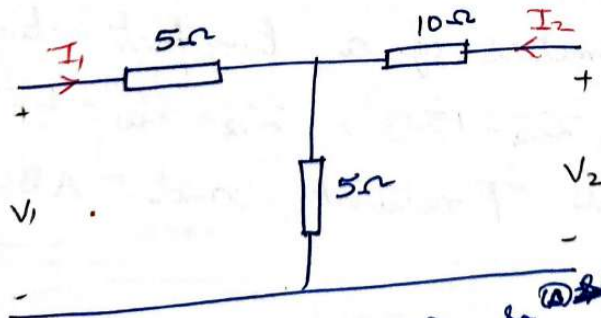
$$Z_{22} = Z_b + Z_c = 15$$

Solve them

Upon solving $Z_c = 5\Omega$

$$Z_a = 10 - 5 = 5\Omega$$

$$Z_b = 15 - 5 = 10\Omega$$



T parameters

$$V_1 = AV_2 + BI_2$$

$$I_1 = CV_2 + DI_2$$

$$V_1 = 10I_1 + 5I_2 \quad \text{--- (1)}$$

$$V_2 = 5I_1 + 15I_2 \quad \text{--- (2)}$$

$$I_1 = \frac{1}{5}V_2 - 3I_2 \quad \text{--- (A)}$$

$$V_1 = 2V_2 - 30I_2 + 5I_2$$

$$V_1 = 2V_2 - 25I_2 \quad \text{--- (B)}$$

$$A = 2, \quad B = 25$$

$$C = \frac{1}{5}, \quad D = 3$$